**Annual Report - General Order No. R5-2007-0035**
*Reporting period 01/01/2023 to 12/31/2023.*

DAIRY FACILITY INFORMATION

**A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY:** Carlos C. Lourenco Dairy

Physical address of dairy:

| Number and Street | City | County | Zip Code |
|---|---|---|---|
| 6557 Oak AVE | Merced | Merced | 95340 |

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 10/01/1995

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

0056-0200-0018-0000

**B. OPERATORS**

Lourenco, Carlos

Operator name: Lourenco, Carlos

Telephone no.: (209) 495-8654 (Landline) (Cellular)

| Mailing Address Number and Street | City | State | Zip Code |
|---|---|---|---|
| 6557 Oak AVE | Merced | CA | 95341 |

**This operator is responsible for paying permit fees.**

**C. OWNERS**

Lourenco, Carlos

Legal owner name: Lourenco, Carlos

Telephone no.: (209) 495-8654 (Landline) (Cellular)

| Mailing Address Number and Street | City | State | Zip Code |
|---|---|---|---|
| 6557 Oak AVE | Merced | CA | 95341 |

**This owner is responsible for paying permit fees.**

AVAILABLE NUTRIENTS

## A. HERD INFORMATION

| | Milk Cows | Dry Cows | Bred Heifers (15-24 mo.) | Heifers (7-14 mo. to breeding) | Calves (4-6 mo.) | Calves (0-3 mo.) |
|---|---|---|---|---|---|---|
| Number open confinement | 0 | 0 | 0 | 0 | 0 | 0 |
| Number under roof | 1,090 | 150 | 310 | 275 | 235 | 0 |
| Maximum number | 1,110 | 162 | 322 | 283 | 241 | 0 |
| Average number | 1,090 | 150 | 310 | 275 | 235 | 0 |
| Avg live weight (lbs) | 1,400 | 1,450 | 1,000 | 750 | | |

Predominant milk cow breed: Holstein

Average milk production: 68 *pounds per cow per day*

## B. MANURE GENERATED

Total manure excreted by the herd: 36,598.65 *tons per reporting period*

Total nitrogen from manure: 457,119.19 *lbs per reporting period*

After ammonia losses (30% loss applied): 319,983.43 *lbs per reporting period*

Total phosphorus from manure: 75,130.54 *lbs per reporting period*

Total potassium from manure: 207,312.80 *lbs per reporting period*

Total salt from manure: 547,719.00 *lbs per reporting period*

## C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 31,932,000 *gallons*

Total nitrogen generated: 89,892.87 *lbs*

Total phosphorus generated: 13,850.88 *lbs*

Total potassium generated: 112,705.57 *lbs*

Total salt generated: 748,208.03 *lbs*

| | | |
|---|---|---|
| | 31,932,000 | *gallons applied* |
| + | 0 | *gallons exported* |
| - | 0 | *gallons imported* |
| = | 31,932,000 | *gallons generated* |

## D. FRESH WATER SOURCES

| Source Description | Type |
|---|---|
| D-1,2 | Ground water |
| D-3 | Ground water |
| DW-2 | Ground water |
| MID Canal | Surface water |
| Well #6-1 | Ground water |

| Source Description | Type |
|---|---|
| Well #7-1 | Ground water |
| Well #8 | Ground water |

## E. SUBSURFACE (TILE) DRAINAGE SOURCES

*No subsurface (tile) drainage sources entered.*

## F. NUTRIENT IMPORTS

*No dry manure nutrient imports entered.*

*No process wastewater nutrient imports entered.*

*No commercial or other nutrient imports entered.*

## G. NUTRIENT EXPORTS

*No solid nutrient exports entered.*

*No liquid nutrient exports entered.*

APPLICATION AREA

## A. LIST OF LAND APPLICATION AREAS

| Field name | Controlled acres | Cropable acres | Total harvests | Type of waste applied | Parcel number |
|---|---|---|---|---|---|
| Field 1 | 106 | 106 | 8 | process wastewater | 0056-0020-0018-0000 |
| Field 10 | 96 | 96 | 2 | manure | 0065-0070-0015-0000 |
| Field 11 | 29 | 29 | 4 | process wastewater | 0056-0200-0010-0000 |
| Field 2 | 47 | 47 | 2 | process wastewater | 0056-0020-0018-0000 |
| Field 234 | 113 | 113 | 1 | process wastewater | 0056-0020-0018-0000 |
| Field 3 | 35 | 35 | 2 | process wastewater | 0056-0020-0018-0000 |
| Field 4 | 30 | 30 | 2 | process wastewater | 0056-0020-0018-0000 |
| Field 5 | 58 | 58 | 2 | manure | 0065-0700-0043-0000<br>0065-0700-0044-0000 |
| Field 6 | 26 | 26 | 2 | manure | 0065-0700-0040-0000 |
| Field 7 | 72 | 72 | 8 | none | 0065-0700-0018-0000<br>0065-0700-0019-0000<br>0065-0700-0020-0000<br>0065-0700-0021-0000<br>0065-0700-0031-0000<br>0065-0700-0032-0000 |
| Field 8 | 75 | 75 | 2 | manure | 0065-0700-0024-0000 |
| Field 9 | 40 | 40 | 8 | none | 0065-0700-0050-0000 |
| Totals for areas that were used for application | 615 | 615 | 27 | | |
| Totals for areas that were not used for application | 112 | 112 | 16 | | |
| Land application area totals | 727 | 727 | 43 | | |

## B. CROPS AND HARVESTS

Field 1

Field name: Field 1

## Field 1

### 10/14/2021: Alfalfa, hay

Crop: Alfalfa, hay  Acres planted: 106  Plant date: 10/14/2021

| Harvest date | Yield | Reporting basis | Density (lbs/cu ft) | Moisture (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 04/14/2023 | 302.00 *ton* | Dry-weight | | 60.4 | 37,200.00 | 3,000.00 | 47,000.00 | | 13.10 |
| 05/10/2023 | 130.00 *ton* | Dry-weight | | 7.5 | 38,000.00 | 3,800.00 | 37,000.00 | | 11.70 |
| 06/06/2023 | 107.00 *ton* | Dry-weight | | 7.5 | 38,000.00 | 3,800.00 | 37,000.00 | | 11.70 |
| 07/09/2023 | 198.00 *ton* | Dry-weight | | 7.5 | 38,000.00 | 3,800.00 | 37,000.00 | | 11.70 |
| 08/11/2023 | 226.00 *ton* | Dry-weight | | 7.9 | 34,600.00 | 3,700.00 | 37,000.00 | | 11.70 |
| 09/14/2023 | 158.00 *ton* | Dry-weight | | 7.9 | 34,600.00 | 3,700.00 | 37,000.00 | | 11.70 |
| 10/12/2023 | 88.00 *ton* | Dry-weight | | 7.9 | 34,600.00 | 3,700.00 | 37,000.00 | | 11.70 |
| 11/13/2023 | 155.00 *ton* | Dry-weight | | 60.4 | 37,200.00 | 3,000.00 | 47,000.00 | | 13.10 |

| | Yield (tons/acre) | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Salt (lbs/acre) |
|---|---|---|---|---|---|
| Anticipated harvest content | 8.00 | 480.00 | 43.20 | 336.00 | 1,480.00 |
| Total actual harvest content | 12.87 | 699.31 | 69.44 | 744.87 | 2,295.22 |

## Field 10

Field name: Field 10

### 10/11/2022: Wheat, silage, soft dough

Crop: Wheat, silage, soft dough  Acres planted: 96  Plant date: 10/11/2022

| Harvest date | Yield | Reporting basis | Density (lbs/cu ft) | Moisture (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 05/23/2023 | 1,387.00 *ton* | Dry-weight | | 56.6 | 13,500.00 | 2,110.00 | 10,500.00 | | 8.45 |

| | Yield (tons/acre) | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Salt (lbs/acre) |
|---|---|---|---|---|---|
| Anticipated harvest content | 18.00 | 198.00 | 30.60 | 149.40 | 1,080.00 |
| Total actual harvest content | 14.45 | 169.30 | 26.46 | 131.68 | 1,059.70 |

## Field 10

### 06/15/2023: Corn, silage

Crop: Corn, silage  Acres planted: 96  Plant date: 06/15/2023

| Harvest date | Yield | Reporting basis | Density (lbs/cu ft) | Moisture (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 09/29/2023 | 2,073.00 *ton* | Dry-weight | | 66.4 | 9,600.00 | 2,050.00 | 14,600.00 | | 6.60 |

| | Yield (tons/acre) | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Salt (lbs/acre) |
|---|---|---|---|---|---|
| Anticipated harvest content | 28.00 | 252.00 | 42.00 | 184.80 | 1,120.00 |
| Total actual harvest content | 21.59 | 139.31 | 29.75 | 211.86 | 957.73 |

## Field 11

Field name: Field 11

### 10/14/2022: Wheat, silage, soft dough

Crop: Wheat, silage, soft dough  Acres planted: 29  Plant date: 10/14/2022

| Harvest date | Yield | Reporting basis | Density (lbs/cu ft) | Moisture (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 04/18/2023 | 504.00 *ton* | Dry-weight | | 72.0 | 20,400.00 | 5,100.00 | 57,800.00 | | 13.20 |

| | Yield (tons/acre) | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Salt (lbs/acre) |
|---|---|---|---|---|---|
| Anticipated harvest content | 18.00 | 198.00 | 30.60 | 149.40 | 1,080.00 |
| Total actual harvest content | 17.38 | 198.54 | 49.64 | 562.53 | 1,284.68 |

## Field 11

### 05/02/2023: Sudangrass, silage

Crop: Sudangrass, silage  Acres planted: 29  Plant date: 05/02/2023

| Harvest date | Yield | Reporting basis | Density (lbs/cu ft) | Moisture (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 07/12/2023 | 370.00 *ton* | Dry-weight | | 74.6 | 18,500.00 | 3,190.00 | 29,100.00 | | 13.30 |
| 08/21/2023 | 507.00 *ton* | Dry-weight | | 74.6 | 18,500.00 | 3,190.00 | 29,100.00 | | 13.30 |
| 09/29/2023 | 136.00 *ton* | Dry-weight | | 74.6 | 18,500.00 | 3,190.00 | 29,100.00 | | 13.30 |

| | Yield (tons/acre) | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Salt (lbs/acre) |
|---|---|---|---|---|---|
| Anticipated harvest content | 28.00 | 308.00 | 47.60 | 336.00 | 1,260.00 |
| Total actual harvest content | 34.93 | 328.28 | 56.61 | 516.38 | 2,360.08 |

## Field 2

Field name: Field 2

### 10/14/2022: Wheat, silage, soft dough

Crop: Wheat, silage, soft dough  Acres planted: 47  Plant date: 10/14/2022

| Harvest date | Yield | Reporting basis | Density (lbs/cu ft) | Moisture (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 04/18/2023 | 692.00 *ton* | Dry-weight | | 68.4 | 20,600.00 | 5,300.00 | 64,900.00 | | 14.70 |

| | Yield (tons/acre) | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Salt (lbs/acre) |
|---|---|---|---|---|---|
| Anticipated harvest content | 18.00 | 198.00 | 30.60 | 149.40 | 1,080.00 |
| Total actual harvest content | 14.72 | 191.69 | 49.32 | 603.91 | 1,367.86 |

## Field 2

### 05/12/2023: Corn, silage

Crop: Corn, silage  Acres planted: 47  Plant date: 05/12/2023

| Harvest date | Yield | Reporting basis | Density (lbs/cu ft) | Moisture (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 08/24/2023 | 1,208.00 *ton* | Dry-weight | | 68.2 | 9,300.00 | 1,290.00 | 17,900.00 | | 6.27 |

| | Yield (tons/acre) | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Salt (lbs/acre) |
|---|---|---|---|---|---|
| Anticipated harvest content | 28.00 | 252.00 | 42.00 | 184.80 | 1,120.00 |
| Total actual harvest content | 25.70 | 152.02 | 21.09 | 292.60 | 1,024.93 |

## Field 234

Field name: Field 234

### 08/30/2023: Sudangrass, silage

Crop: Sudangrass, silage  Acres planted: 113  Plant date: 08/30/2023

| Harvest date | Yield | Reporting basis | Density (lbs/cu ft) | Moisture (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 10/24/2023 | 1,568.00 *ton* | Dry-weight | | 81.6 | 18,100.00 | 3,060.00 | 30,400.00 | | 24.60 |

| | Yield (tons/acre) | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Salt (lbs/acre) |
|---|---|---|---|---|---|
| Anticipated harvest content | 12.00 | 132.00 | 20.40 | 144.00 | 540.00 |
| Total actual harvest content | 13.88 | 92.43 | 15.63 | 155.23 | 1,256.18 |

## Field 3

Field name: Field 3

## Field 3

### 10/14/2022: Wheat, silage, soft dough

Crop: Wheat, silage, soft dough  Acres planted: 35  Plant date: 10/14/2022

| Harvest date | Yield | Reporting basis | Density (lbs/cu ft) | Moisture (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 04/18/2023 | 516.00 *ton* | Dry-weight | | 68.6 | 20,400.00 | 4,900.00 | 60,700.00 | | 14.60 |

| | Yield (tons/acre) | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Salt (lbs/acre) |
|---|---|---|---|---|---|
| Anticipated harvest content | 18.00 | 198.00 | 30.60 | 149.40 | 1,080.00 |
| Total actual harvest content | 14.74 | 188.87 | 45.37 | 561.99 | 1,351.74 |

### 05/12/2023: Corn, silage

Crop: Corn, silage  Acres planted: 35  Plant date: 05/12/2023

| Harvest date | Yield | Reporting basis | Density (lbs/cu ft) | Moisture (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 08/24/2023 | 899.00 *ton* | Dry-weight | | 69.2 | 10,300.00 | 2,810.00 | 14,400.00 | | 6.80 |

| | Yield (tons/acre) | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Salt (lbs/acre) |
|---|---|---|---|---|---|
| Anticipated harvest content | 28.00 | 252.00 | 42.00 | 184.80 | 1,120.00 |
| Total actual harvest content | 25.69 | 162.97 | 44.46 | 227.84 | 1,075.92 |

## Field 4

Field name: Field 4

### 10/14/2022: Wheat, silage, soft dough

Crop: Wheat, silage, soft dough  Acres planted: 30  Plant date: 10/14/2022

| Harvest date | Yield | Reporting basis | Density (lbs/cu ft) | Moisture (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 04/18/2023 | 442.00 *ton* | Dry-weight | | 70.4 | 18,300.00 | 5,300.00 | 64,100.00 | | 13.70 |

| | Yield (tons/acre) | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Salt (lbs/acre) |
|---|---|---|---|---|---|
| Anticipated harvest content | 18.00 | 198.00 | 30.60 | 149.40 | 1,080.00 |
| Total actual harvest content | 14.73 | 159.62 | 46.23 | 559.09 | 1,194.93 |

## Field 4

### 05/12/2023: Corn, silage

Crop: Corn, silage  Acres planted: 30  Plant date: 05/12/2023

| Harvest date | Yield | Reporting basis | Density (lbs/cu ft) | Moisture (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 08/25/2023 | 771.00 *ton* | Dry-weight | | 69.0 | 9,600.00 | 2,890.00 | 14,600.00 | | 6.67 |

| | Yield (tons/acre) | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Salt (lbs/acre) |
|---|---|---|---|---|---|
| Anticipated harvest content | 28.00 | 252.00 | 42.00 | 184.80 | 1,120.00 |
| Total actual harvest content | 25.70 | 152.97 | 46.05 | 232.64 | 1,062.80 |

## Field 5

Field name: Field 5

### 11/28/2022: Wheat, silage, soft dough

Crop: Wheat, silage, soft dough  Acres planted: 58  Plant date: 11/28/2022

| Harvest date | Yield | Reporting basis | Density (lbs/cu ft) | Moisture (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 05/22/2023 | 1,088.00 *ton* | Dry-weight | | 58.0 | 13,500.00 | 2,380.00 | 11,200.00 | | 7.47 |

| | Yield (tons/acre) | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Salt (lbs/acre) |
|---|---|---|---|---|---|
| Anticipated harvest content | 18.00 | 198.00 | 30.60 | 149.40 | 1,080.00 |
| Total actual harvest content | 18.76 | 212.72 | 37.50 | 176.48 | 1,177.07 |

### 06/20/2023: Corn, silage

Crop: Corn, silage  Acres planted: 58  Plant date: 06/20/2023

| Harvest date | Yield | Reporting basis | Density (lbs/cu ft) | Moisture (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 09/28/2023 | 1,436.00 *ton* | Dry-weight | | 72.6 | 10,700.00 | 2,690.00 | 12,100.00 | | 3.90 |

| | Yield (tons/acre) | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Salt (lbs/acre) |
|---|---|---|---|---|---|
| Anticipated harvest content | 28.00 | 252.00 | 42.00 | 184.80 | 1,120.00 |
| Total actual harvest content | 24.76 | 145.17 | 36.50 | 164.17 | 529.14 |

## Field 6

Field name: Field 6

### 11/28/2022: Wheat, silage, soft dough

Crop: Wheat, silage, soft dough  Acres planted: 26  Plant date: 11/28/2022

| Harvest date | Yield | Reporting basis | Density (lbs/cu ft) | Moisture (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 05/22/2023 | 738.00 *ton* | Dry-weight | | 58.2 | 14,500.00 | 2,280.00 | 10,800.00 | | 7.74 |

| | Yield (tons/acre) | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Salt (lbs/acre) |
|---|---|---|---|---|---|
| Anticipated harvest content | 18.00 | 198.00 | 30.60 | 149.40 | 1,080.00 |
| Total actual harvest content | 28.38 | 344.08 | 54.10 | 256.28 | 1,836.67 |

### 06/19/2023: Corn, silage

Crop: Corn, silage  Acres planted: 26  Plant date: 06/19/2023

| Harvest date | Yield | Reporting basis | Density (lbs/cu ft) | Moisture (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 09/28/2023 | 919.00 *ton* | Dry-weight | | 70.4 | 11,600.00 | 2,710.00 | 12,400.00 | | 4.70 |

| | Yield (tons/acre) | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Salt (lbs/acre) |
|---|---|---|---|---|---|
| Anticipated harvest content | 28.00 | 252.00 | 42.00 | 184.80 | 1,120.00 |
| Total actual harvest content | 35.35 | 242.73 | 56.71 | 259.47 | 983.47 |

## Field 7

Field name: Field 7

## Field 7

### 10/18/2020: Alfalfa, hay

Crop: Alfalfa, hay  Acres planted: 72  Plant date: 10/18/2020

| Harvest date | Yield | Reporting basis | Density (lbs/cu ft) | Moisture (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 04/18/2023 | 356.00 *ton* | Dry-weight | | 60.4 | 37,200.00 | 3,000.00 | 47,000.00 | | 13.10 |
| 05/09/2023 | 85.00 *ton* | Dry-weight | | 7.8 | 45,700.00 | 3,400.00 | 20,000.00 | | 9.08 |
| 06/05/2023 | 84.00 *ton* | Dry-weight | | 7.8 | 45,700.00 | 3,400.00 | 20,000.00 | | 9.08 |
| 07/08/2023 | 130.00 *ton* | Dry-weight | | 7.8 | 42,900.00 | 3,700.00 | 25,000.00 | | 8.98 |
| 08/10/2023 | 148.00 *ton* | Dry-weight | | 7.8 | 42,900.00 | 3,700.00 | 25,000.00 | | 8.98 |
| 09/13/2023 | 103.00 *ton* | Dry-weight | | 7.8 | 42,900.00 | 3,700.00 | 25,000.00 | | 8.98 |
| 10/11/2023 | 53.00 *ton* | Dry-weight | | 7.8 | 42,900.00 | 3,700.00 | 25,000.00 | | 8.98 |
| 11/13/2023 | 105.00 *ton* | Dry-weight | | 60.4 | 37,200.00 | 3,000.00 | 47,000.00 | | 13.10 |

| | Yield (tons/acre) | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Salt (lbs/acre) |
|---|---|---|---|---|---|
| Anticipated harvest content | 8.00 | 480.00 | 43.20 | 336.00 | 1,480.00 |
| Total actual harvest content | 14.78 | 863.29 | 71.06 | 602.78 | 2,055.46 |

## Field 8

Field name: Field 8

### 10/11/2022: Wheat, silage, soft dough

Crop: Wheat, silage, soft dough  Acres planted: 75  Plant date: 10/11/2022

| Harvest date | Yield | Reporting basis | Density (lbs/cu ft) | Moisture (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 05/23/2023 | 1,375.00 *ton* | Dry-weight | | 55.6 | 19,900.00 | 2,140.00 | 10,800.00 | | 9.11 |

| | Yield (tons/acre) | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Salt (lbs/acre) |
|---|---|---|---|---|---|
| Anticipated harvest content | 18.00 | 198.00 | 30.60 | 149.40 | 1,080.00 |
| Total actual harvest content | 18.33 | 323.97 | 34.84 | 175.82 | 1,483.11 |

## Field 8

### 06/16/2023: Corn, silage

Crop: Corn, silage  Acres planted: 75  Plant date: 06/16/2023

| Harvest date | Yield | Reporting basis | Density (lbs/cu ft) | Moisture (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 09/29/2023 | 2,141.00 *ton* | Dry-weight | | 64.2 | 9,700.00 | 2,000.00 | 10,700.00 | | 5.10 |

| | Yield (tons/acre) | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Salt (lbs/acre) |
|---|---|---|---|---|---|
| Anticipated harvest content | 28.00 | 252.00 | 42.00 | 184.80 | 1,120.00 |
| Total actual harvest content | 28.55 | 198.26 | 40.88 | 218.70 | 1,042.41 |

## Field 9

Field name: Field 9

### 10/19/2020: Alfalfa, hay

Crop: Alfalfa, hay  Acres planted: 40  Plant date: 10/19/2020

| Harvest date | Yield | Reporting basis | Density (lbs/cu ft) | Moisture (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 04/18/2023 | 198.00 *ton* | Dry-weight | | 60.4 | 37,200.00 | 3,000.00 | 47,000.00 | | 13.10 |
| 05/09/2023 | 46.00 *ton* | Dry-weight | | 7.8 | 45,700.00 | 3,400.00 | 20,000.00 | | 9.08 |
| 06/05/2023 | 47.00 *ton* | Dry-weight | | 7.8 | 45,700.00 | 3,400.00 | 20,000.00 | | 9.08 |
| 07/08/2023 | 72.00 *ton* | Dry-weight | | 7.8 | 45,700.00 | 3,400.00 | 20,000.00 | | 9.08 |
| 08/10/2023 | 82.00 *ton* | Dry-weight | | 7.8 | 42,900.00 | 3,700.00 | 25,000.00 | | 8.98 |
| 09/13/2023 | 57.00 *ton* | Dry-weight | | 7.8 | 42,900.00 | 3,700.00 | 25,000.00 | | 8.98 |
| 10/11/2023 | 29.00 *ton* | Dry-weight | | 7.8 | 42,900.00 | 3,700.00 | 25,000.00 | | 8.98 |
| 11/13/2023 | 58.00 *ton* | Dry-weight | | 60.4 | 37,200.00 | 3,000.00 | 47,000.00 | | 13.10 |

| | Yield (tons/acre) | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Salt (lbs/acre) |
|---|---|---|---|---|---|
| Anticipated harvest content | 8.00 | 480.00 | 43.20 | 336.00 | 1,480.00 |
| Total actual harvest content | 14.73 | 868.43 | 69.72 | 583.98 | 2,050.17 |

NUTRIENT BUDGET

## A. LAND APPLICATIONS

Field 1 - 10/14/2021: Alfalfa, hay

Field name: Field 1

Crop: Alfalfa, hay

Plant date: 10/14/2021

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 05/22/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Blended Lg and Fresh Water Application | Process wastewater | 75.03 | 11.62 | 97.51 | 477.16 | 2,196,000.00 *gal* |
| MID Canal | Surface water | 0.00 | 0.00 | 0.00 | 213.13 | 11,520,000.00 *gal* |
| Application event totals | | 75.03 | 11.62 | 97.51 | 690.29 | |

| | | | | |
|---|---|---|---|---|
| 06/12/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| MID Canal | Surface water | 0.00 | 0.00 | 0.00 | 217.57 | 11,760,000.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 217.57 | |

| | | | | |
|---|---|---|---|---|
| 06/26/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| DW-2 | Ground water | 1.69 | 0.00 | 0.00 | 326.49 | 11,520,000.00 *gal* |
| Application event totals | | 1.69 | 0.00 | 0.00 | 326.49 | |

| | | | | |
|---|---|---|---|---|
| 07/14/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| MID Canal | Surface water | 0.00 | 0.00 | 0.00 | 204.25 | 11,040,000.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 204.25 | |

| | | | | |
|---|---|---|---|---|
| 07/28/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| MID Canal | Surface water | 0.00 | 0.00 | 0.00 | 199.81 | 10,800,000.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 199.81 | |

## Field 1 - 10/14/2021: Alfalfa, hay

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 08/16/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| MID Canal | Surface water | 0.00 | 0.00 | 0.00 | 187.82 | 10,152,000.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 187.82 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 08/28/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| MID Canal | Surface water | 0.00 | 0.00 | 0.00 | 190.93 | 10,320,000.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 190.93 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 09/21/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| MID Canal | Surface water | 0.00 | 0.00 | 0.00 | 201.14 | 10,872,000.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 201.14 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 10/18/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| MID Canal | Surface water | 0.00 | 0.00 | 0.00 | 195.37 | 10,560,000.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 195.37 | |

## Field 10 - 10/11/2022: Wheat, silage, soft dough

Field name: Field 10

Crop: Wheat, silage, soft dough

Plant date: 10/11/2022

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 09/20/2022 | Broadcast/incorporate | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Manure Application | Corral solids | 212.46 | 54.85 | 243.10 | 3,755.16 | 864.00 *ton* |
| Application event totals | | 212.46 | 54.85 | 243.10 | 3,755.16 | |

## Field 10 - 10/11/2022: Wheat, silage, soft dough

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 10/30/2022 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Well #8 | Ground water | 0.30 | 0.00 | 0.00 | 37.96 | 10,650,000.00 *gal* |
| Application event totals | | 0.30 | 0.00 | 0.00 | 37.96 | |

| | | | | |
|---|---|---|---|---|
| 04/25/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| MID Canal | Surface water | 0.00 | 0.00 | 0.00 | 183.85 | 9,000,000.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 183.85 | |

## Field 10 - 06/15/2023: Corn, silage

Field name: Field 10

Crop: Corn, silage

Plant date: 06/15/2023

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 05/29/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| MID Canal | Surface water | 0.00 | 0.00 | 0.00 | 336.45 | 16,470,000.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 336.45 | |

| | | | | |
|---|---|---|---|---|
| 06/01/2023 | Broadcast/incorporate | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Manure Application | Corral solids | 179.12 | 25.35 | 91.86 | 0.00 | 1,460.00 *ton* |
| Application event totals | | 179.12 | 25.35 | 91.86 | 0.00 | |

| | | | | |
|---|---|---|---|---|
| 07/19/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Well #7-1 | Ground water | 2.73 | 0.00 | 0.00 | 312.62 | 9,720,000.00 *gal* |
| Application event totals | | 2.73 | 0.00 | 0.00 | 312.62 | |

## Field 10 - 06/15/2023: Corn, silage

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 07/30/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Well #8 | Ground water | 0.18 | 0.00 | 0.00 | 23.09 | 6,480,000.00 *gal* |
| Application event totals | | 0.18 | 0.00 | 0.00 | 23.09 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 08/10/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| MID Canal | Surface water | 0.00 | 0.00 | 0.00 | 121.34 | 5,940,000.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 121.34 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 08/22/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| MID Canal | Surface water | 0.00 | 0.00 | 0.00 | 191.39 | 9,369,000.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 191.39 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 08/30/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| MID Canal | Surface water | 0.00 | 0.00 | 0.00 | 196.90 | 9,639,000.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 196.90 | |

## Field 11 - 10/14/2022: Wheat, silage, soft dough

Field name: Field 11

Crop: Wheat, silage, soft dough

Plant date: 10/14/2022

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 10/24/2022 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| DW-2 | Ground water | 1.20 | 0.00 | 0.00 | 231.84 | 2,238,000.00 *gal* |
| Application event totals | | 1.20 | 0.00 | 0.00 | 231.84 | |

## Field 11 - 10/14/2022: Wheat, silage, soft dough

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 12/19/2022 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Blended Lg and Fresh Water Application | Process wastewater | 86.27 | 19.49 | 139.23 | 678.74 | 864,000.00 *gal* |
| DW-2 | Ground water | 1.34 | 0.00 | 0.00 | 259.19 | 2,502,000.00 *gal* |
| Application event totals | | 87.61 | 19.49 | 139.23 | 937.93 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 03/15/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Blended Lg and Fresh Water Application | Process wastewater | 130.53 | 18.65 | 183.48 | 1,335.11 | 1,296,000.00 *gal* |
| Application event totals | | 130.53 | 18.65 | 183.48 | 1,335.11 | |

## Field 11 - 05/02/2023: Sudangrass, silage

Field name: Field 11

Crop: Sudangrass, silage

Plant date: 05/02/2023

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 05/20/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Blended Lg and Fresh Water Application | Process wastewater | 53.95 | 8.35 | 70.11 | 343.10 | 432,000.00 *gal* |
| DW-2 | Ground water | 1.83 | 0.00 | 0.00 | 354.29 | 3,420,000.00 *gal* |
| Application event totals | | 55.78 | 8.35 | 70.11 | 697.39 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 06/14/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Blended Lg and Fresh Water Application | Process wastewater | 53.95 | 8.35 | 70.11 | 343.10 | 432,000.00 *gal* |
| DW-2 | Ground water | 1.78 | 0.00 | 0.00 | 344.97 | 3,330,000.00 *gal* |
| Application event totals | | 55.73 | 8.35 | 70.11 | 688.07 | |

Field 11 - 05/02/2023: Sudangrass, silage

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 06/28/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Blended Lg and Fresh Water Application | Process wastewater | 53.95 | 8.35 | 70.11 | 343.10 | 432,000.00 *gal* |
| MID Canal | Surface water | 0.00 | 0.00 | 0.00 | 164.32 | 2,430,000.00 *gal* |
| Application event totals | | 53.95 | 8.35 | 70.11 | 507.42 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 07/13/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Blended Lg and Fresh Water Application | Process wastewater | 37.67 | 1.23 | 20.14 | 443.79 | 648,000.00 *gal* |
| MID Canal | Surface water | 0.00 | 0.00 | 0.00 | 176.50 | 2,610,000.00 *gal* |
| Application event totals | | 37.67 | 1.23 | 20.14 | 620.29 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 07/27/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Blended Lg and Fresh Water Application | Process wastewater | 37.67 | 1.23 | 20.14 | 443.79 | 648,000.00 *gal* |
| MID Canal | Surface water | 0.00 | 0.00 | 0.00 | 176.50 | 2,610,000.00 *gal* |
| Application event totals | | 37.67 | 1.23 | 20.14 | 620.29 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 08/05/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Blended Lg and Fresh Water Application | Process wastewater | 63.77 | 13.35 | 72.54 | 518.38 | 648,000.00 *gal* |
| DW-2 | Ground water | 1.30 | 0.00 | 0.00 | 251.73 | 2,430,000.00 *gal* |
| Application event totals | | 65.07 | 13.35 | 72.54 | 770.11 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 08/23/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Blended Lg and Fresh Water Application | Process wastewater | 63.77 | 13.35 | 72.54 | 518.38 | 648,000.00 *gal* |
| DW-2 | Ground water | 1.06 | 0.00 | 0.00 | 205.11 | 1,980,000.00 *gal* |
| Application event totals | | 64.83 | 13.35 | 72.54 | 723.49 | |

## Field 11 - 05/02/2023: Sudangrass, silage

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 09/06/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| DW-2 | Ground water | 1.54 | 0.00 | 0.00 | 298.35 | 2,880,000.00 *gal* |
| Application event totals | | 1.54 | 0.00 | 0.00 | 298.35 | |

## Field 2 - 10/14/2022: Wheat, silage, soft dough

Field name: Field 2

Crop: Wheat, silage, soft dough

Plant date: 10/14/2022

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 10/26/2022 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| DW-2 | Ground water | 1.72 | 0.00 | 0.00 | 333.66 | 5,220,000.00 *gal* |
| Application event totals | | 1.72 | 0.00 | 0.00 | 333.66 | |

| | | | | |
|---|---|---|---|---|
| 12/21/2022 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Blended Lg and Fresh Water Application | Process wastewater | 137.52 | 31.07 | 221.93 | 1,081.90 | 2,232,000.00 *gal* |
| DW-2 | Ground water | 1.23 | 0.00 | 0.00 | 237.78 | 3,720,000.00 *gal* |
| Application event totals | | 138.74 | 31.07 | 221.93 | 1,319.67 | |

| | | | | |
|---|---|---|---|---|
| 03/20/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Blended Lg and Fresh Water Application | Process wastewater | 134.23 | 19.18 | 188.69 | 1,372.98 | 2,160,000.00 *gal* |
| Application event totals | | 134.23 | 19.18 | 188.69 | 1,372.98 | |

## Field 2 - 05/12/2023: Corn, silage

Field name: Field 2

Crop: Corn, silage

Plant date: 05/12/2023

Field 2 - 05/12/2023: Corn, silage

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 04/25/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Blended Lg and Fresh Water Application | Process wastewater | 155.35 | 24.05 | 201.88 | 987.93 | 2,016,000.00 *gal* |
| DW-2 | Ground water | 1.40 | 0.00 | 0.00 | 270.38 | 4,230,000.00 *gal* |
| Application event totals | | 156.75 | 24.05 | 201.88 | 1,258.31 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 06/19/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| MID Canal | Surface water | 0.00 | 0.00 | 0.00 | 236.58 | 5,670,000.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 236.58 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 06/29/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| MID Canal | Surface water | 0.00 | 0.00 | 0.00 | 180.25 | 4,320,000.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 180.25 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 07/11/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Blended Lg and Fresh Water Application | Process wastewater | 46.48 | 1.52 | 24.85 | 547.66 | 1,296,000.00 *gal* |
| MID Canal | Surface water | 0.00 | 0.00 | 0.00 | 172.74 | 4,140,000.00 *gal* |
| Application event totals | | 46.48 | 1.52 | 24.85 | 720.40 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 07/23/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| DW-2 | Ground water | 1.46 | 0.00 | 0.00 | 281.88 | 4,410,000.00 *gal* |
| Application event totals | | 1.46 | 0.00 | 0.00 | 281.88 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 08/03/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| MID Canal | Surface water | 0.00 | 0.00 | 0.00 | 172.74 | 4,140,000.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 172.74 | |

## Field 234 - 08/30/2023: Sudangrass, silage

Field name: Field 234

Crop: Sudangrass, silage — Plant date: 08/30/2023

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 08/30/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Blended Lg and Fresh Water Application | Process wastewater | 42.96 | 1.40 | 22.97 | 506.19 | 2,880,000.00 *gal* |
| MID Canal | Surface water | 0.00 | 0.00 | 0.00 | 109.57 | 6,313,500.00 *gal* |
| Application event totals | | 42.96 | 1.40 | 22.97 | 615.76 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 09/19/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Blended Lg and Fresh Water Application | Process wastewater | 34.37 | 1.12 | 18.38 | 404.96 | 2,304,000.00 *gal* |
| MID Canal | Surface water | 0.00 | 0.00 | 0.00 | 85.91 | 4,950,000.00 *gal* |
| Application event totals | | 34.37 | 1.12 | 18.38 | 490.86 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 10/02/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Blended Lg and Fresh Water Application | Process wastewater | 43.64 | 9.14 | 49.64 | 354.76 | 1,728,000.00 *gal* |
| MID Canal | Surface water | 0.00 | 0.00 | 0.00 | 81.69 | 4,707,000.00 *gal* |
| Application event totals | | 43.64 | 9.14 | 49.64 | 436.45 | |

## Field 3 - 10/14/2022: Wheat, silage, soft dough

Field name: Field 3

Crop: Wheat, silage, soft dough — Plant date: 10/14/2022

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 10/26/2022 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Blended Lg and Fresh Water Application | Process wastewater | 71.48 | 16.15 | 115.36 | 562.39 | 864,000.00 *gal* |
| DW-2 | Ground water | 1.01 | 0.00 | 0.00 | 194.67 | 2,268,000.00 *gal* |
| Application event totals | | 72.49 | 16.15 | 115.36 | 757.06 | |

## Field 3 - 10/14/2022: Wheat, silage, soft dough

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 12/21/2022 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Blended Lg and Fresh Water Application | Process wastewater | 107.22 | 24.23 | 173.04 | 843.58 | 1,296,000.00 *gal* |
| DW-2 | Ground water | 0.99 | 0.00 | 0.00 | 191.58 | 2,232,000.00 *gal* |
| Application event totals | | 108.21 | 24.23 | 173.04 | 1,035.16 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 01/15/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Blended Lg and Fresh Water Application | Process wastewater | 72.10 | 10.30 | 101.35 | 737.49 | 864,000.00 *gal* |
| Application event totals | | 72.10 | 10.30 | 101.35 | 737.49 | |

## Field 3 - 05/12/2023: Corn, silage

Field name: Field 3

Crop: Corn, silage

Plant date: 05/12/2023

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 04/25/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Blended Lg and Fresh Water Application | Process wastewater | 223.51 | 34.61 | 290.46 | 1,421.42 | 2,160,000.00 *gal* |
| MID Canal | Surface water | 0.00 | 0.00 | 0.00 | 205.41 | 3,666,000.00 *gal* |
| Application event totals | | 223.51 | 34.61 | 290.46 | 1,626.82 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 06/19/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| MID Canal | Surface water | 0.00 | 0.00 | 0.00 | 275.33 | 4,914,000.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 275.33 | |

## Field 3 - 05/12/2023: Corn, silage

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 06/29/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| MID Canal | Surface water | 0.00 | 0.00 | 0.00 | 209.78 | 3,744,000.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 209.78 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 07/11/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| DW-2 | Ground water | 1.59 | 0.00 | 0.00 | 307.97 | 3,588,000.00 *gal* |
| Application event totals | | 1.59 | 0.00 | 0.00 | 307.97 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 07/23/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| MID Canal | Surface water | 0.00 | 0.00 | 0.00 | 214.15 | 3,822,000.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 214.15 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 08/03/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| MID Canal | Surface water | 0.00 | 0.00 | 0.00 | 201.04 | 3,588,000.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 201.04 | |

## Field 4 - 10/14/2022: Wheat, silage, soft dough

Field name: Field 4

Crop: Wheat, silage, soft dough

Plant date: 10/14/2022

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 10/26/2022 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Blended Lg and Fresh Water Application | Process wastewater | 83.40 | 18.84 | 134.59 | 656.12 | 864,000.00 *gal* |
| DW-2 | Ground water | 0.98 | 0.00 | 0.00 | 189.26 | 1,890,000.00 *gal* |
| Application event totals | | 84.37 | 18.84 | 134.59 | 845.38 | |

## Field 4 - 10/14/2022: Wheat, silage, soft dough

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 12/21/2022 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| DW-2 | Ground water | 0.96 | 0.00 | 0.00 | 186.26 | 1,860,000.00 *gal* |
| Application event totals | | 0.96 | 0.00 | 0.00 | 186.26 | |

| | | | | |
|---|---|---|---|---|
| 02/20/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Blended Lg and Fresh Water Application | Process wastewater | 126.18 | 18.03 | 177.37 | 1,290.60 | 1,296,000.00 *gal* |
| Application event totals | | 126.18 | 18.03 | 177.37 | 1,290.60 | |

## Field 4 - 05/12/2023: Corn, silage

Field name: Field 4

Crop: Corn, silage

Plant date: 05/12/2023

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 04/25/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Blended Lg and Fresh Water Application | Process wastewater | 208.61 | 32.30 | 271.10 | 1,326.65 | 1,728,000.00 *gal* |
| MID Canal | Surface water | 0.00 | 0.00 | 0.00 | 108.25 | 1,656,000.00 *gal* |
| Application event totals | | 208.61 | 32.30 | 271.10 | 1,434.91 | |

| | | | | |
|---|---|---|---|---|
| 06/19/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| DW-2 | Ground water | 2.35 | 0.00 | 0.00 | 454.24 | 4,536,000.00 *gal* |
| Application event totals | | 2.35 | 0.00 | 0.00 | 454.24 | |

| | | | | |
|---|---|---|---|---|
| 06/29/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| MID Canal | Surface water | 0.00 | 0.00 | 0.00 | 225.92 | 3,456,000.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 225.92 | |

## Field 4 - 05/12/2023: Corn, silage

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 07/11/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| MID Canal | Surface water | 0.00 | 0.00 | 0.00 | 216.50 | 3,312,000.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 216.50 | |

| | | | | |
|---|---|---|---|---|
| 07/23/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| MID Canal | Surface water | 0.00 | 0.00 | 0.00 | 230.62 | 3,528,000.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 230.62 | |

| | | | | |
|---|---|---|---|---|
| 08/03/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| MID Canal | Surface water | 0.00 | 0.00 | 0.00 | 216.50 | 3,312,000.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 216.50 | |

## Field 5 - 11/28/2022: Wheat, silage, soft dough

Field name: Field 5

Crop: Wheat, silage, soft dough

Plant date: 11/28/2022

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 09/15/2022 | Broadcast/incorporate | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Manure Application | Corral solids | 244.21 | 63.04 | 279.42 | 4,316.28 | 600.00 *ton* |
| Application event totals | | 244.21 | 63.04 | 279.42 | 4,316.28 | |

| | | | | |
|---|---|---|---|---|
| 10/25/2022 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Well #6-1 | Ground water | 20.74 | 0.00 | 0.00 | 1,071.77 | 10,080,000.00 *gal* |
| Application event totals | | 20.74 | 0.00 | 0.00 | 1,071.77 | |

## Field 5 - 11/28/2022: Wheat, silage, soft dough

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 04/26/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| MID Canal | Surface water | 0.00 | 0.00 | 0.00 | 365.17 | 10,800,000.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 365.17 | |

## Field 5 - 06/20/2023: Corn, silage

Field name: Field 5

Crop: Corn, silage

Plant date: 06/20/2023

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 06/02/2023 | Broadcast/incorporate | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Manure Application | Corral solids | 166.94 | 47.68 | 256.07 | 0.00 | 780.00 *ton* |
| Application event totals | | 166.94 | 47.68 | 256.07 | 0.00 | |

| | | | | |
|---|---|---|---|---|
| 06/04/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| MID Canal | Surface water | 0.00 | 0.00 | 0.00 | 426.03 | 12,600,000.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 426.03 | |

| | | | | |
|---|---|---|---|---|
| 07/19/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Well #6-1 | Ground water | 23.89 | 0.00 | 0.00 | 1,234.77 | 11,613,000.00 *gal* |
| Application event totals | | 23.89 | 0.00 | 0.00 | 1,234.77 | |

| | | | | |
|---|---|---|---|---|
| 07/30/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Well #6-1 | Ground water | 22.17 | 0.00 | 0.00 | 1,145.46 | 10,773,000.00 *gal* |
| Application event totals | | 22.17 | 0.00 | 0.00 | 1,145.46 | |

## Field 5 - 06/20/2023: Corn, silage

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 08/11/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| MID Canal | Surface water | 0.00 | 0.00 | 0.00 | 288.99 | 8,547,000.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 288.99 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 08/23/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| MID Canal | Surface water | 0.00 | 0.00 | 0.00 | 227.21 | 6,720,000.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 227.21 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 09/05/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Well #6-1 | Ground water | 13.83 | 0.00 | 0.00 | 714.52 | 6,720,000.00 *gal* |
| Application event totals | | 13.83 | 0.00 | 0.00 | 714.52 | |

## Field 6 - 11/28/2022: Wheat, silage, soft dough

Field name: Field 6

Crop: Wheat, silage, soft dough

Plant date: 11/28/2022

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 09/16/2022 | Broadcast/incorporate | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Manure Application | Corral solids | 272.39 | 70.31 | 311.66 | 4,814.31 | 300.00 *ton* |
| Application event totals | | 272.39 | 70.31 | 311.66 | 4,814.31 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 10/24/2022 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Well #6-1 | Ground water | 16.63 | 0.00 | 0.00 | 859.22 | 3,622,500.00 *gal* |
| Application event totals | | 16.63 | 0.00 | 0.00 | 859.22 | |

## Field 6 - 11/28/2022: Wheat, silage, soft dough

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 04/25/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| MID Canal | Surface water | 0.00 | 0.00 | 0.00 | 271.53 | 3,600,000.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 271.53 | |

## Field 6 - 06/19/2023: Corn, silage

Field name: Field 6

Crop: Corn, silage

Plant date: 06/19/2023

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 06/01/2023 | Broadcast/incorporate | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Manure Application | Corral solids | 248.27 | 70.90 | 380.82 | 0.00 | 520.00 *ton* |
| Application event totals | | 248.27 | 70.90 | 380.82 | 0.00 | |

| | | | | |
|---|---|---|---|---|
| 06/03/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| MID Canal | Surface water | 0.00 | 0.00 | 0.00 | 260.22 | 3,450,000.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 260.22 | |

| | | | | |
|---|---|---|---|---|
| 07/18/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| MID Canal | Surface water | 0.00 | 0.00 | 0.00 | 237.59 | 3,150,000.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 237.59 | |

| | | | | |
|---|---|---|---|---|
| 07/29/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Well #6-1 | Ground water | 11.02 | 0.00 | 0.00 | 569.26 | 2,400,000.00 *gal* |
| Application event totals | | 11.02 | 0.00 | 0.00 | 569.26 | |

## Field 6 - 06/19/2023: Corn, silage

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 08/10/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Well #6-1 | Ground water | 10.53 | 0.00 | 0.00 | 544.35 | 2,295,000.00 *gal* |
| Application event totals | | 10.53 | 0.00 | 0.00 | 544.35 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 08/22/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| MID Canal | Surface water | 0.00 | 0.00 | 0.00 | 169.71 | 2,250,000.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 169.71 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 09/04/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Well #6-1 | Ground water | 9.64 | 0.00 | 0.00 | 498.10 | 2,100,000.00 *gal* |
| Application event totals | | 9.64 | 0.00 | 0.00 | 498.10 | |

## Field 7 - 10/18/2020: Alfalfa, hay

Field name: Field 7

Crop: Alfalfa, hay

Plant date: 10/18/2020

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 05/23/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| MID Canal | Surface water | 0.00 | 0.00 | 0.00 | 264.75 | 9,720,000.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 264.75 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 06/09/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| MID Canal | Surface water | 0.00 | 0.00 | 0.00 | 264.75 | 9,720,000.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 264.75 | |

Field 7 - 10/18/2020: Alfalfa, hay

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 06/24/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Well #7-1 | Ground water | 3.54 | 0.00 | 0.00 | 405.25 | 9,450,000.00 *gal* |
| Application event totals | | 3.54 | 0.00 | 0.00 | 405.25 | |

| | | | | |
|---|---|---|---|---|
| 07/13/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| MID Canal | Surface water | 0.00 | 0.00 | 0.00 | 143.77 | 5,278,500.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 143.77 | |

| | | | | |
|---|---|---|---|---|
| 07/28/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| MID Canal | Surface water | 0.00 | 0.00 | 0.00 | 220.62 | 8,100,000.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 220.62 | |

| | | | | |
|---|---|---|---|---|
| 08/16/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Well #7-1 | Ground water | 2.43 | 0.00 | 0.00 | 277.89 | 6,480,000.00 *gal* |
| Application event totals | | 2.43 | 0.00 | 0.00 | 277.89 | |

| | | | | |
|---|---|---|---|---|
| 08/30/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| MID Canal | Surface water | 0.00 | 0.00 | 0.00 | 154.43 | 5,670,000.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 154.43 | |

| | | | | |
|---|---|---|---|---|
| 09/20/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| MID Canal | Surface water | 0.00 | 0.00 | 0.00 | 176.50 | 6,480,000.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 176.50 | |

## Field 7 - 10/18/2020: Alfalfa, hay

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 10/18/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Well #7-1 | Ground water | 2.76 | 0.00 | 0.00 | 316.10 | 7,371,000.00 *gal* |
| Application event totals | | 2.76 | 0.00 | 0.00 | 316.10 | |

## Field 8 - 10/11/2022: Wheat, silage, soft dough

Field name: Field 8

Crop: Wheat, silage, soft dough

Plant date: 10/11/2022

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 09/17/2022 | Broadcast/incorporate | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Manure Application | Corral solids | 270.06 | 69.71 | 309.00 | 4,773.23 | 858.00 *ton* |
| Application event totals | | 270.06 | 69.71 | 309.00 | 4,773.23 | |

| | | | | |
|---|---|---|---|---|
| 10/27/2022 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Well #7-1 | Ground water | 6.47 | 0.00 | 0.00 | 741.41 | 18,009,000.00 *gal* |
| Application event totals | | 6.47 | 0.00 | 0.00 | 741.41 | |

| | | | | |
|---|---|---|---|---|
| 04/27/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Well #8 | Ground water | 0.58 | 0.00 | 0.00 | 73.90 | 16,200,000.00 *gal* |
| Application event totals | | 0.58 | 0.00 | 0.00 | 73.90 | |

## Field 8 - 06/16/2023: Corn, silage

Field name: Field 8

Crop: Corn, silage

Plant date: 06/16/2023

## Field 8 - 06/16/2023: Corn, silage

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 05/26/2023 | Broadcast/incorporate | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Manure Application | Corral solids | 223.44 | 63.81 | 342.74 | 0.00 | 1,350.00 *ton* |
| Application event totals | | 223.44 | 63.81 | 342.74 | 0.00 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 05/29/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| MID Canal | Surface water | 0.00 | 0.00 | 0.00 | 435.59 | 16,659,000.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 435.59 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 07/19/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| MID Canal | Surface water | 0.00 | 0.00 | 0.00 | 289.45 | 11,070,000.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 289.45 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 07/30/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Well #8 | Ground water | 0.09 | 0.00 | 0.00 | 11.95 | 2,619,000.00 *gal* |
| Application event totals | | 0.09 | 0.00 | 0.00 | 11.95 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 08/10/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| MID Canal | Surface water | 0.00 | 0.00 | 0.00 | 232.98 | 8,910,000.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 232.98 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 08/22/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| MID Canal | Surface water | 0.00 | 0.00 | 0.00 | 317.69 | 12,150,000.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 317.69 | |

## Field 8 - 06/16/2023: Corn, silage

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 08/30/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| MID Canal | Surface water | 0.00 | 0.00 | 0.00 | 117.90 | 4,509,000.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 117.90 | |

## Field 9 - 10/19/2020: Alfalfa, hay

Field name: Field 9

Crop: Alfalfa, hay

Plant date: 10/19/2020

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 05/20/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| MID Canal | Surface water | 0.00 | 0.00 | 0.00 | 317.69 | 6,480,000.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 317.69 | |

| | | | | |
|---|---|---|---|---|
| 06/09/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Well #7-1 | Ground water | 4.55 | 0.00 | 0.00 | 521.04 | 6,750,000.00 *gal* |
| Application event totals | | 4.55 | 0.00 | 0.00 | 521.04 | |

| | | | | |
|---|---|---|---|---|
| 06/24/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| MID Canal | Surface water | 0.00 | 0.00 | 0.00 | 321.67 | 6,561,000.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 321.67 | |

| | | | | |
|---|---|---|---|---|
| 07/13/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Well #7-1 | Ground water | 4.55 | 0.00 | 0.00 | 521.04 | 6,750,000.00 *gal* |
| Application event totals | | 4.55 | 0.00 | 0.00 | 521.04 | |

## Field 9 - 10/19/2020: Alfalfa, hay

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 07/28/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| MID Canal | Surface water | 0.00 | 0.00 | 0.00 | 321.67 | 6,561,000.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 321.67 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 08/16/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| MID Canal | Surface water | 0.00 | 0.00 | 0.00 | 317.69 | 6,480,000.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 317.69 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 08/30/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| MID Canal | Surface water | 0.00 | 0.00 | 0.00 | 238.27 | 4,860,000.00 *gal* |
| Application event totals | | 0.00 | 0.00 | 0.00 | 238.27 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 09/20/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Well #7-1 | Ground water | 4.00 | 0.00 | 0.00 | 458.52 | 5,940,000.00 *gal* |
| Application event totals | | 4.00 | 0.00 | 0.00 | 458.52 | |

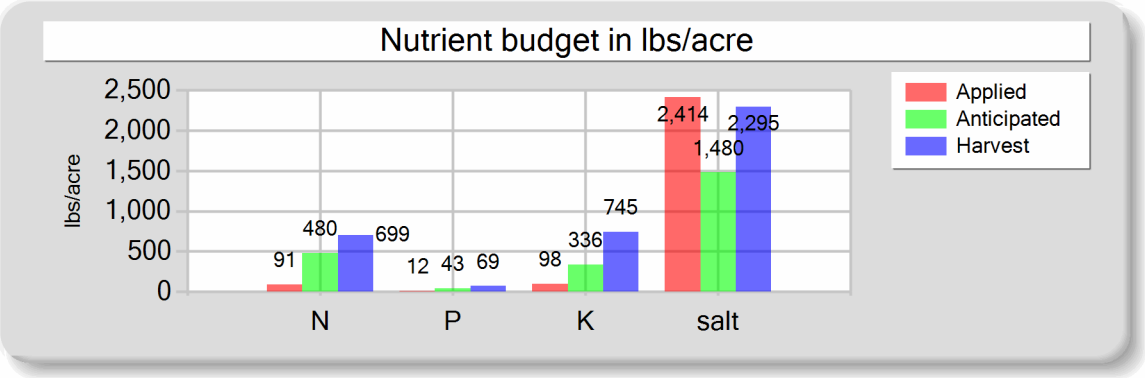
| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 10/18/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Well #8 | Ground water | 0.18 | 0.00 | 0.00 | 3.34 | 2,670,000.00 *gal* |
| Well #7-1 | Ground water | 2.02 | 0.00 | 0.00 | 231.57 | 3,000,000.00 *gal* |
| Application event totals | | 2.20 | 0.00 | 0.00 | 234.92 | |

## B. NUTRIENT BUDGET

Field 1 - 10/14/2021: Alfalfa, hay

Field name: Field 1 Crop: Alfalfa, hay Plant date: 10/14/2021




| | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Total salt (lbs/acre) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 0.00 | 0.00 | 0.00 | 0.00 |
| Dry manure | 0.00 | 0.00 | 0.00 | 0.00 |
| Process wastewater | 75.03 | 11.62 | 97.51 | 477.16 |
| Fresh water | 1.69 | 0.00 | 0.00 | 1,936.50 |
| Atmospheric deposition | 14.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 90.72 | 11.62 | 97.51 | 2,413.66 |
| Anticipated crop nutrient removal | 480.00 | 43.20 | 336.00 | 1,480.00 |
| Actual crop nutrient removal | 699.31 | 69.44 | 744.87 | 2,295.22 |
| Nutrient balance | -608.59 | -57.82 | -647.36 | 118.44 |
| Applied to removed ratio | 0.13 | 0.17 | 0.13 | 1.05 |

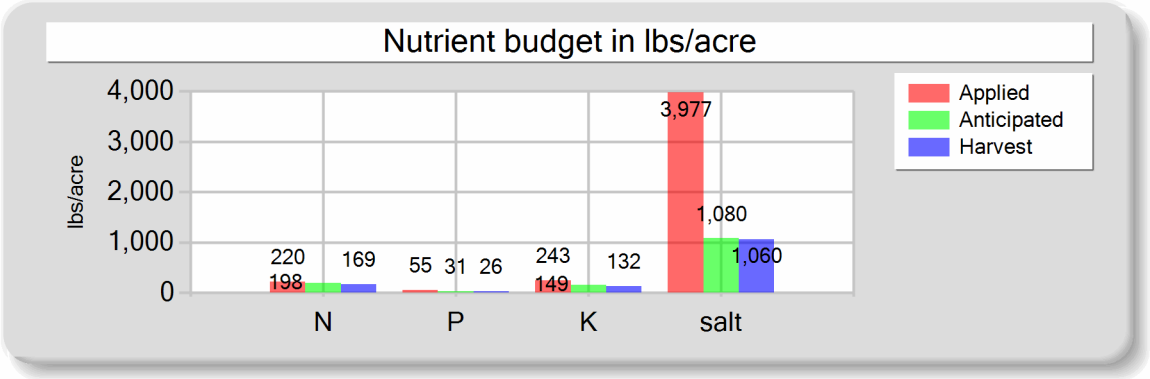
| Fresh water applied | |
|---|---|
| 98,544,000.00 | *gallons* |
| 3,629.04 | *acre-inches* |
| 34.24 | *inches/acre* |

| Process wastewater applied | |
|---|---|
| 2,196,000.00 | *gallons* |
| 80.87 | *acre-inches* |
| 0.76 | *inches/acre* |

| Total harvests for the crop | |
|---|---|
| 8 | *harvests* |

## Field 10 - 10/11/2022: Wheat, silage, soft dough

Field name: Field 10 Crop: Wheat, silage, soft dough Plant date: 10/11/2022




| | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Total salt (lbs/acre) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 0.00 | 0.00 | 0.00 | 0.00 |
| Dry manure | 212.46 | 54.85 | 243.10 | 3,755.16 |
| Process wastewater | 0.00 | 0.00 | 0.00 | 0.00 |
| Fresh water | 0.30 | 0.00 | 0.00 | 221.81 |
| Atmospheric deposition | 7.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 219.76 | 54.85 | 243.10 | 3,976.97 |
| Anticipated crop nutrient removal | 198.00 | 30.60 | 149.40 | 1,080.00 |
| Actual crop nutrient removal | 169.30 | 26.46 | 131.68 | 1,059.70 |
| Nutrient balance | 50.46 | 28.38 | 111.42 | 2,917.27 |
| Applied to removed ratio | 1.30 | 2.07 | 1.85 | 3.75 |

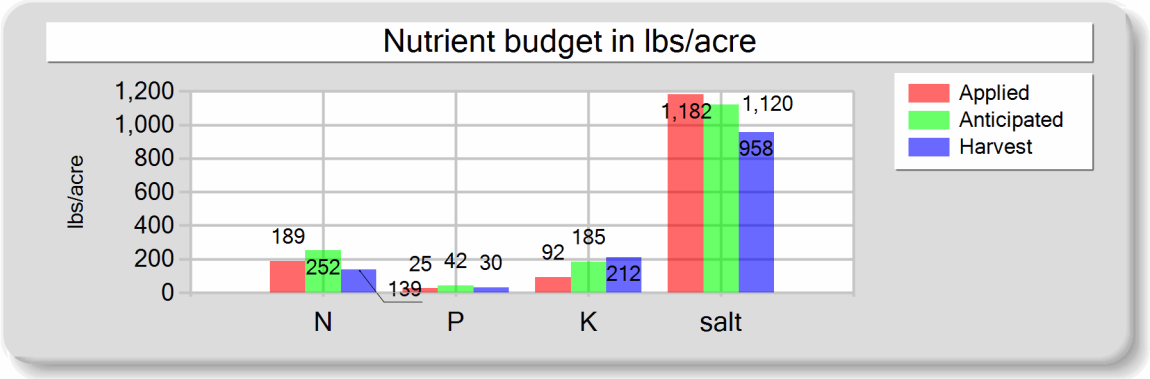
| Fresh water applied | |
|---|---|
| 19,650,000.00 | *gallons* |
| 723.64 | *acre-inches* |
| 7.54 | *inches/acre* |

| Process wastewater applied | |
|---|---|
| 0.00 | *gallons* |
| 0.00 | *acre-inches* |
| 0.00 | *inches/acre* |

| Total harvests for the crop | |
|---|---|
| 1 | *harvests* |

## Field 10 - 06/15/2023: Corn, silage

Field name: Field 10 Crop: Corn, silage Plant date: 06/15/2023




| | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Total salt (lbs/acre) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 0.00 | 0.00 | 0.00 | 0.00 |
| Dry manure | 179.12 | 25.35 | 91.86 | 0.00 |
| Process wastewater | 0.00 | 0.00 | 0.00 | 0.00 |
| Fresh water | 2.91 | 0.00 | 0.00 | 1,181.80 |
| Atmospheric deposition | 7.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 189.03 | 25.35 | 91.86 | 1,181.80 |
| Anticipated crop nutrient removal | 252.00 | 42.00 | 184.80 | 1,120.00 |
| Actual crop nutrient removal | 139.31 | 29.75 | 211.86 | 957.73 |
| Nutrient balance | 49.73 | -4.39 | -120.00 | 224.07 |
| Applied to removed ratio | 1.36 | 0.85 | 0.43 | 1.23 |

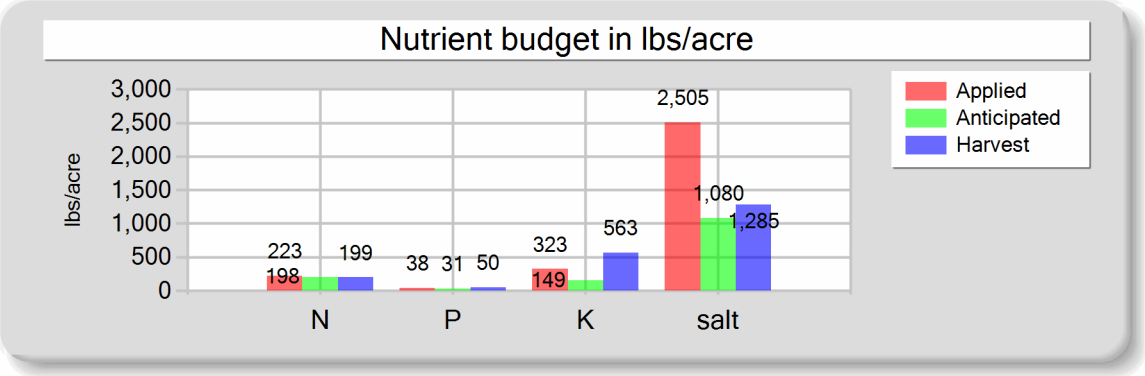
| Fresh water applied | |
|---|---|
| 57,618,000.00 | *gallons* |
| 2,121.88 | *acre-inches* |
| 22.10 | *inches/acre* |

| Process wastewater applied | |
|---|---|
| 0.00 | *gallons* |
| 0.00 | *acre-inches* |
| 0.00 | *inches/acre* |

| Total harvests for the crop | |
|---|---|
| 1 | *harvests* |

## Field 11 - 10/14/2022: Wheat, silage, soft dough

Field name: Field 11 Crop: Wheat, silage, soft dough Plant date: 10/14/2022




| | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Total salt (lbs/acre) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 0.00 | 0.00 | 0.00 | 0.00 |
| Dry manure | 0.00 | 0.00 | 0.00 | 0.00 |
| Process wastewater | 216.80 | 38.14 | 322.71 | 2,013.85 |
| Fresh water | 2.54 | 0.00 | 0.00 | 491.03 |
| Atmospheric deposition | 3.50 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 222.84 | 38.14 | 322.71 | 2,504.88 |
| Anticipated crop nutrient removal | 198.00 | 30.60 | 149.40 | 1,080.00 |
| Actual crop nutrient removal | 198.54 | 49.64 | 562.53 | 1,284.68 |
| Nutrient balance | 24.30 | -11.50 | -239.82 | 1,220.20 |
| Applied to removed ratio | 1.12 | 0.77 | 0.57 | 1.95 |

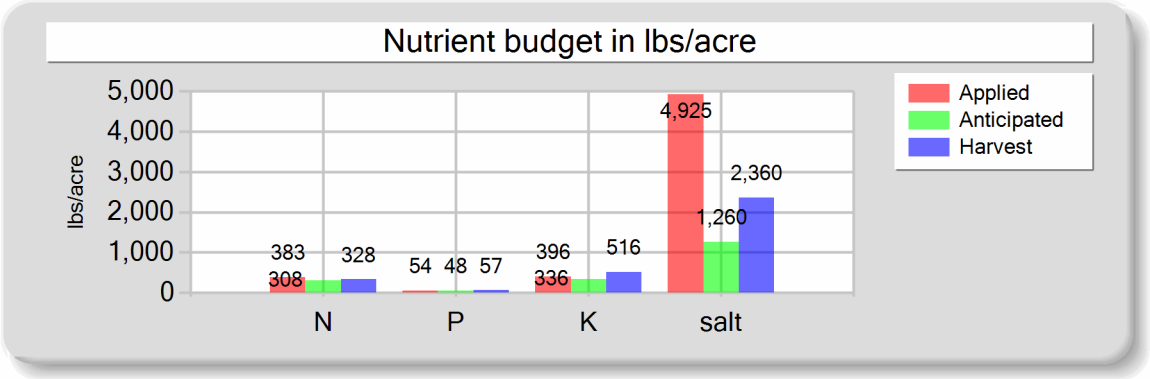
| Fresh water applied | |
|---|---|
| 4,740,000.00 | *gallons* |
| 174.56 | *acre-inches* |
| 6.02 | *inches/acre* |

| Process wastewater applied | |
|---|---|
| 2,160,000.00 | *gallons* |
| 79.55 | *acre-inches* |
| 2.74 | *inches/acre* |

| Total harvests for the crop | |
|---|---|
| 1 | *harvests* |

## Field 11 - 05/02/2023: Sudangrass, silage

Field name: Field 11 Crop: Sudangrass, silage Plant date: 05/02/2023




| | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Total salt (lbs/acre) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 0.00 | 0.00 | 0.00 | 0.00 |
| Dry manure | 0.00 | 0.00 | 0.00 | 0.00 |
| Process wastewater | 364.73 | 54.22 | 395.68 | 2,953.65 |
| Fresh water | 7.51 | 0.00 | 0.00 | 1,971.77 |
| Atmospheric deposition | 10.50 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 382.75 | 54.22 | 395.68 | 4,925.41 |
| Anticipated crop nutrient removal | 308.00 | 47.60 | 336.00 | 1,260.00 |
| Actual crop nutrient removal | 328.28 | 56.61 | 516.38 | 2,360.08 |
| Nutrient balance | 54.46 | -2.39 | -120.69 | 2,565.33 |
| Applied to removed ratio | 1.17 | 0.96 | 0.77 | 2.09 |

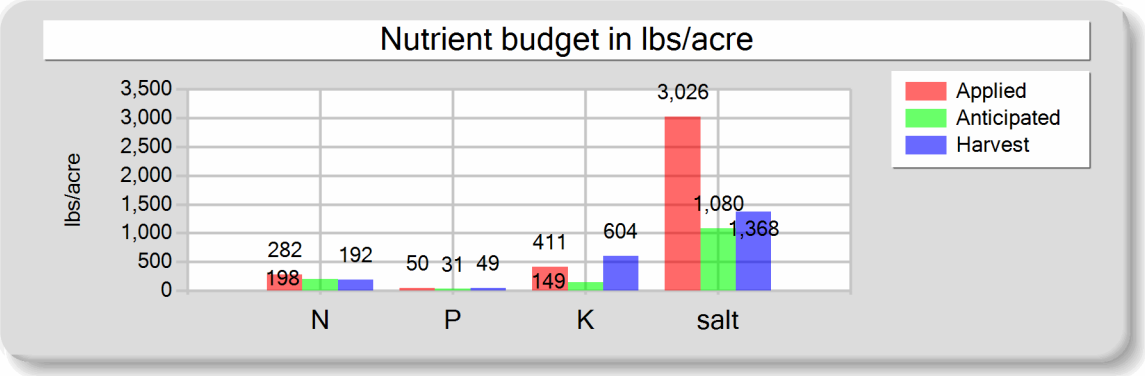
| Fresh water applied | |
|---|---|
| 21,690,000.00 | *gallons* |
| 798.77 | *acre-inches* |
| 27.54 | *inches/acre* |

| Process wastewater applied | |
|---|---|
| 3,888,000.00 | *gallons* |
| 143.18 | *acre-inches* |
| 4.94 | *inches/acre* |

| Total harvests for the crop | |
|---|---|
| 3 | *harvests* |

## Field 2 - 10/14/2022: Wheat, silage, soft dough

Field name: Field 2 Crop: Wheat, silage, soft dough Plant date: 10/14/2022




| | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Total salt (lbs/acre) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 0.00 | 0.00 | 0.00 | 0.00 |
| Dry manure | 0.00 | 0.00 | 0.00 | 0.00 |
| Process wastewater | 271.75 | 50.25 | 410.62 | 2,454.88 |
| Fresh water | 2.95 | 0.00 | 0.00 | 571.44 |
| Atmospheric deposition | 7.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 281.70 | 50.25 | 410.62 | 3,026.32 |
| Anticipated crop nutrient removal | 198.00 | 30.60 | 149.40 | 1,080.00 |
| Actual crop nutrient removal | 191.69 | 49.32 | 603.91 | 1,367.86 |
| Nutrient balance | 90.01 | 0.93 | -193.29 | 1,658.45 |
| Applied to removed ratio | 1.47 | 1.02 | 0.68 | 2.21 |

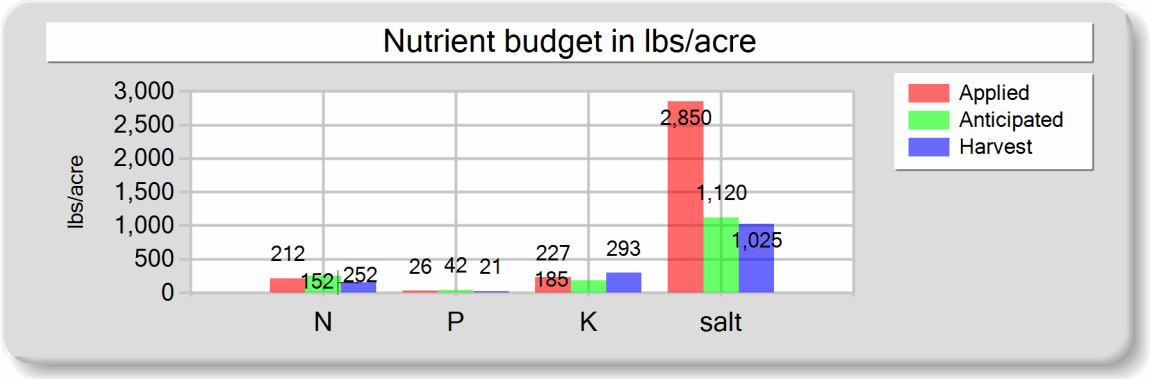
| Fresh water applied | |
|---|---|
| 8,940,000.00 | *gallons* |
| 329.23 | *acre-inches* |
| 7.00 | *inches/acre* |

| Process wastewater applied | |
|---|---|
| 4,392,000.00 | *gallons* |
| 161.74 | *acre-inches* |
| 3.44 | *inches/acre* |

| Total harvests for the crop | |
|---|---|
| 1 | *harvests* |

## Field 2 - 05/12/2023: Corn, silage

Field name: Field 2 Crop: Corn, silage Plant date: 05/12/2023




| | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Total salt (lbs/acre) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 0.00 | 0.00 | 0.00 | 0.00 |
| Dry manure | 0.00 | 0.00 | 0.00 | 0.00 |
| Process wastewater | 201.83 | 25.57 | 226.73 | 1,535.59 |
| Fresh water | 2.85 | 0.00 | 0.00 | 1,314.58 |
| Atmospheric deposition | 7.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 211.68 | 25.57 | 226.73 | 2,850.17 |
| Anticipated crop nutrient removal | 252.00 | 42.00 | 184.80 | 1,120.00 |
| Actual crop nutrient removal | 152.02 | 21.09 | 292.60 | 1,024.93 |
| Nutrient balance | 59.66 | 4.48 | -65.87 | 1,825.24 |
| Applied to removed ratio | 1.39 | 1.21 | 0.77 | 2.78 |

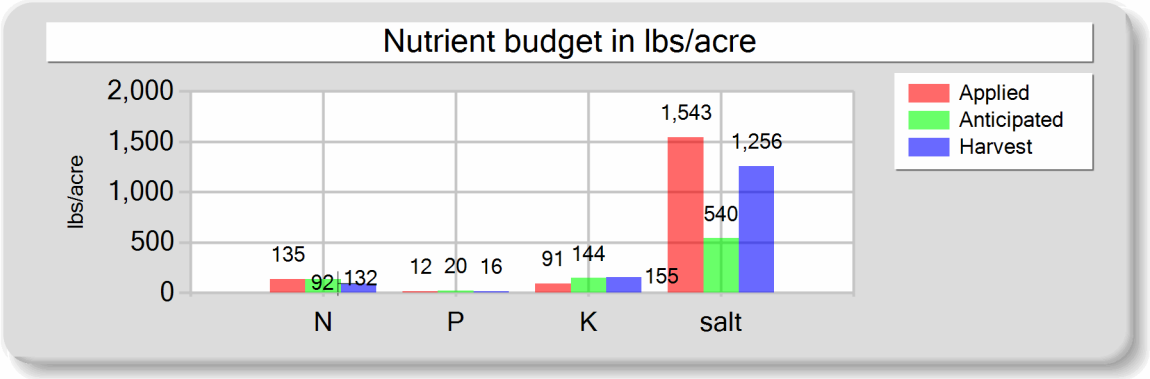
| Fresh water applied | |
|---|---|
| 26,910,000.00 | *gallons* |
| 991.00 | *acre-inches* |
| 21.09 | *inches/acre* |

| Process wastewater applied | |
|---|---|
| 3,312,000.00 | *gallons* |
| 121.97 | *acre-inches* |
| 2.60 | *inches/acre* |

| Total harvests for the crop | |
|---|---|
| 1 | *harvests* |

## Field 234 - 08/30/2023: Sudangrass, silage

Field name: Field 234 Crop: Sudangrass, silage Plant date: 08/30/2023




| | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Total salt (lbs/acre) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 0.00 | 0.00 | 0.00 | 0.00 |
| Dry manure | 0.00 | 0.00 | 0.00 | 0.00 |
| Process wastewater | 120.98 | 11.66 | 90.99 | 1,265.91 |
| Fresh water | 0.00 | 0.00 | 0.00 | 277.16 |
| Atmospheric deposition | 14.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 134.98 | 11.66 | 90.99 | 1,543.07 |
| Anticipated crop nutrient removal | 132.00 | 20.40 | 144.00 | 540.00 |
| Actual crop nutrient removal | 92.43 | 15.63 | 155.23 | 1,256.18 |
| Nutrient balance | 42.55 | -3.97 | -64.25 | 286.90 |
| Applied to removed ratio | 1.46 | 0.75 | 0.59 | 1.23 |

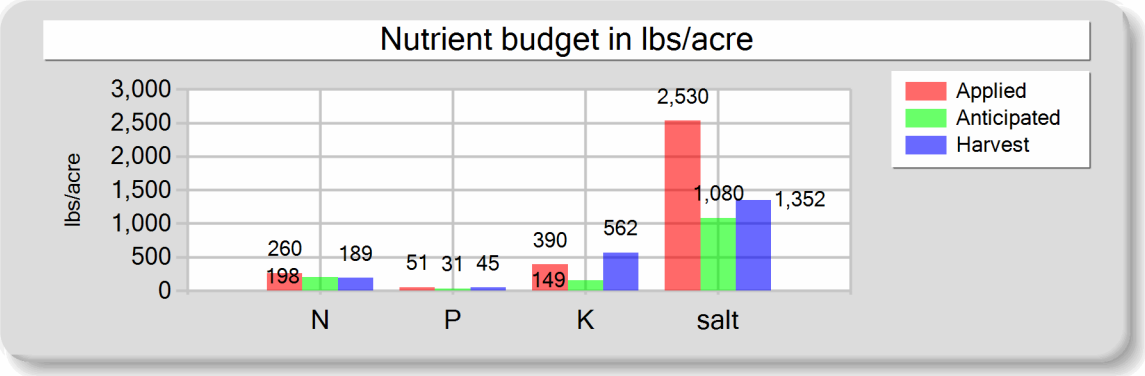
| Fresh water applied | |
|---|---|
| 15,970,500.00 | *gallons* |
| 588.14 | *acre-inches* |
| 5.20 | *inches/acre* |

| Process wastewater applied | |
|---|---|
| 6,912,000.00 | *gallons* |
| 254.55 | *acre-inches* |
| 2.25 | *inches/acre* |

| Total harvests for the crop | |
|---|---|
| 1 | *harvests* |

## Field 3 - 10/14/2022: Wheat, silage, soft dough

Field name: Field 3 Crop: Wheat, silage, soft dough Plant date: 10/14/2022




| | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Total salt (lbs/acre) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 0.00 | 0.00 | 0.00 | 0.00 |
| Dry manure | 0.00 | 0.00 | 0.00 | 0.00 |
| Process wastewater | 250.81 | 50.68 | 389.76 | 2,143.45 |
| Fresh water | 2.00 | 0.00 | 0.00 | 386.25 |
| Atmospheric deposition | 7.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 259.80 | 50.68 | 389.76 | 2,529.71 |
| Anticipated crop nutrient removal | 198.00 | 30.60 | 149.40 | 1,080.00 |
| Actual crop nutrient removal | 188.87 | 45.37 | 561.99 | 1,351.74 |
| Nutrient balance | 70.93 | 5.31 | -172.24 | 1,177.96 |
| Applied to removed ratio | 1.38 | 1.12 | 0.69 | 1.87 |

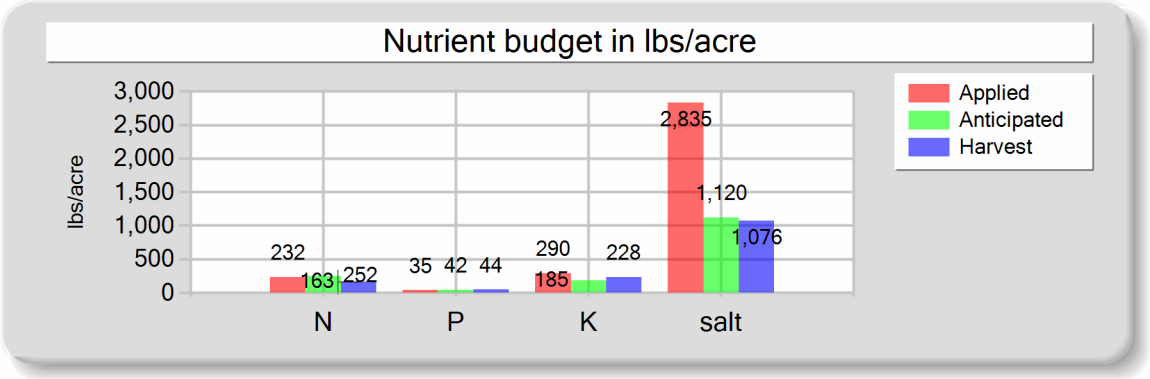
| Fresh water applied | |
|---|---|
| 4,500,000.00 | *gallons* |
| 165.72 | *acre-inches* |
| 4.73 | *inches/acre* |

| Process wastewater applied | |
|---|---|
| 3,024,000.00 | *gallons* |
| 111.36 | *acre-inches* |
| 3.18 | *inches/acre* |

| Total harvests for the crop | |
|---|---|
| 1 | *harvests* |

## Field 3 - 05/12/2023: Corn, silage

Field name: Field 3 Crop: Corn, silage Plant date: 05/12/2023




| | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Total salt (lbs/acre) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 0.00 | 0.00 | 0.00 | 0.00 |
| Dry manure | 0.00 | 0.00 | 0.00 | 0.00 |
| Process wastewater | 223.51 | 34.61 | 290.46 | 1,421.42 |
| Fresh water | 1.59 | 0.00 | 0.00 | 1,413.68 |
| Atmospheric deposition | 7.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 232.10 | 34.61 | 290.46 | 2,835.10 |
| Anticipated crop nutrient removal | 252.00 | 42.00 | 184.80 | 1,120.00 |
| Actual crop nutrient removal | 162.97 | 44.46 | 227.84 | 1,075.92 |
| Nutrient balance | 69.13 | -9.85 | 62.62 | 1,759.18 |
| Applied to removed ratio | 1.42 | 0.78 | 1.27 | 2.64 |

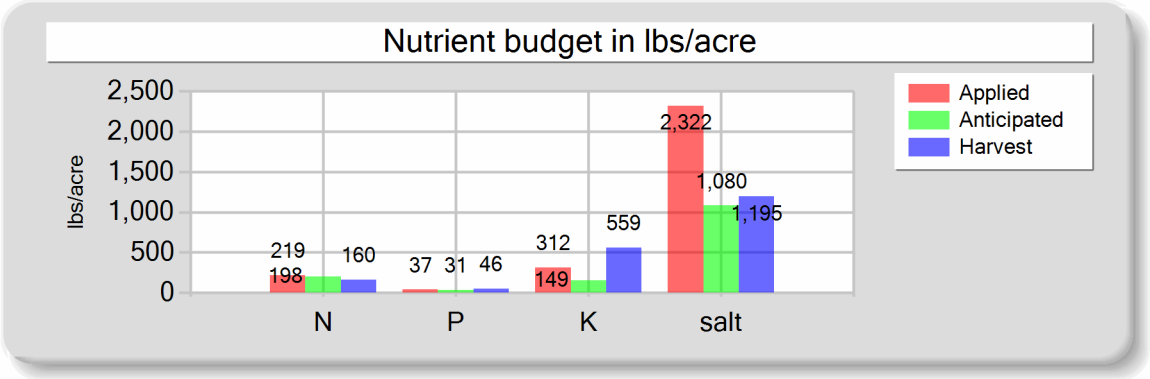
| Fresh water applied | |
|---|---|
| 23,322,000.00 | *gallons* |
| 858.87 | *acre-inches* |
| 24.54 | *inches/acre* |

| Process wastewater applied | |
|---|---|
| 2,160,000.00 | *gallons* |
| 79.55 | *acre-inches* |
| 2.27 | *inches/acre* |

| Total harvests for the crop | |
|---|---|
| 1 | *harvests* |

## Field 4 - 10/14/2022: Wheat, silage, soft dough

Field name: Field 4 Crop: Wheat, silage, soft dough Plant date: 10/14/2022




| | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Total salt (lbs/acre) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 0.00 | 0.00 | 0.00 | 0.00 |
| Dry manure | 0.00 | 0.00 | 0.00 | 0.00 |
| Process wastewater | 209.57 | 36.87 | 311.96 | 1,946.72 |
| Fresh water | 1.94 | 0.00 | 0.00 | 375.53 |
| Atmospheric deposition | 7.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 218.51 | 36.87 | 311.96 | 2,322.25 |
| Anticipated crop nutrient removal | 198.00 | 30.60 | 149.40 | 1,080.00 |
| Actual crop nutrient removal | 159.62 | 46.23 | 559.09 | 1,194.93 |
| Nutrient balance | 58.90 | -9.36 | -247.13 | 1,127.31 |
| Applied to removed ratio | 1.37 | 0.80 | 0.56 | 1.94 |

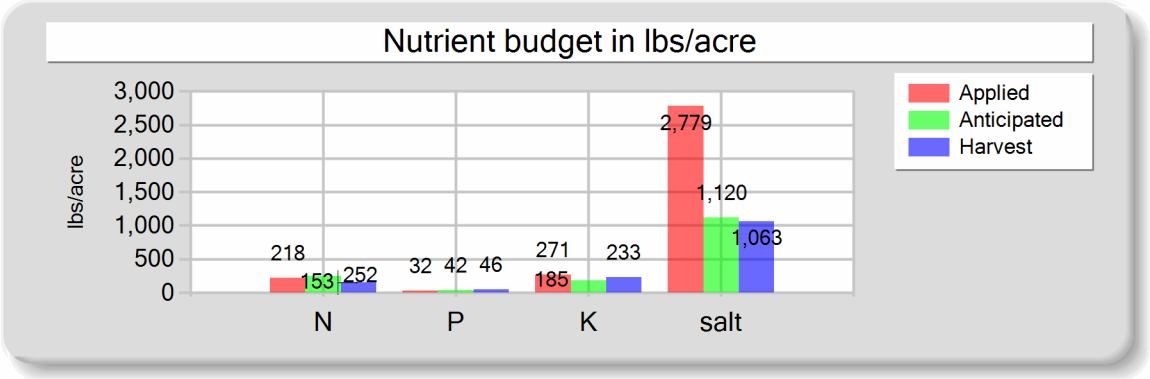
| Fresh water applied | |
|---|---|
| 3,750,000.00 | *gallons* |
| 138.10 | *acre-inches* |
| 4.60 | *inches/acre* |

| Process wastewater applied | |
|---|---|
| 2,160,000.00 | *gallons* |
| 79.55 | *acre-inches* |
| 2.65 | *inches/acre* |

| Total harvests for the crop | |
|---|---|
| 1 | *harvests* |

## Field 4 - 05/12/2023: Corn, silage

Field name: Field 4 Crop: Corn, silage Plant date: 05/12/2023




| | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Total salt (lbs/acre) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 0.00 | 0.00 | 0.00 | 0.00 |
| Dry manure | 0.00 | 0.00 | 0.00 | 0.00 |
| Process wastewater | 208.61 | 32.30 | 271.10 | 1,326.65 |
| Fresh water | 2.35 | 0.00 | 0.00 | 1,452.03 |
| Atmospheric deposition | 7.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 217.96 | 32.30 | 271.10 | 2,778.68 |
| Anticipated crop nutrient removal | 252.00 | 42.00 | 184.80 | 1,120.00 |
| Actual crop nutrient removal | 152.97 | 46.05 | 232.64 | 1,062.80 |
| Nutrient balance | 64.99 | -13.75 | 38.46 | 1,715.89 |
| Applied to removed ratio | 1.42 | 0.70 | 1.17 | 2.61 |

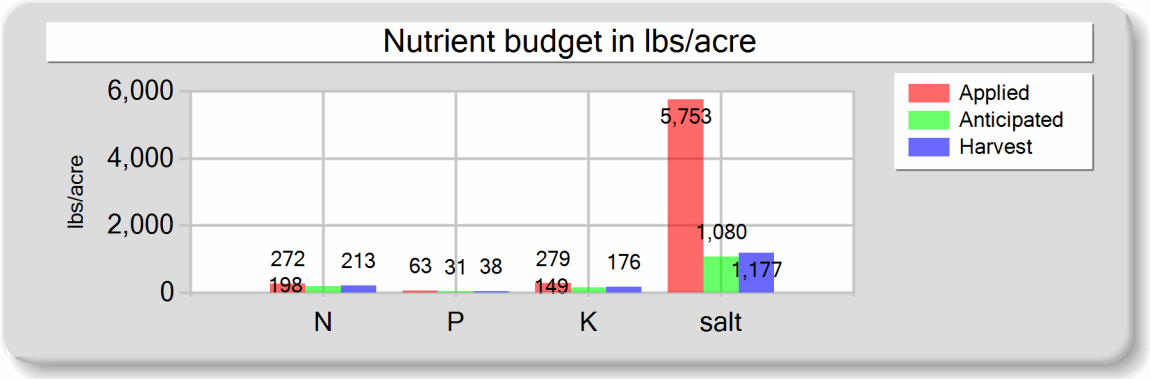
| Fresh water applied | |
|---|---|
| 19,800,000.00 | *gallons* |
| 729.17 | *acre-inches* |
| 24.31 | *inches/acre* |

| Process wastewater applied | |
|---|---|
| 1,728,000.00 | *gallons* |
| 63.64 | *acre-inches* |
| 2.12 | *inches/acre* |

| Total harvests for the crop | |
|---|---|
| 1 | *harvests* |

## Field 5 - 11/28/2022: Wheat, silage, soft dough

Field name: Field 5 Crop: Wheat, silage, soft dough Plant date: 11/28/2022




| | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Total salt (lbs/acre) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 0.00 | 0.00 | 0.00 | 0.00 |
| Dry manure | 244.21 | 63.04 | 279.42 | 4,316.28 |
| Process wastewater | 0.00 | 0.00 | 0.00 | 0.00 |
| Fresh water | 20.74 | 0.00 | 0.00 | 1,436.94 |
| Atmospheric deposition | 7.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 271.95 | 63.04 | 279.42 | 5,753.22 |
| Anticipated crop nutrient removal | 198.00 | 30.60 | 149.40 | 1,080.00 |
| Actual crop nutrient removal | 212.72 | 37.50 | 176.48 | 1,177.07 |
| Nutrient balance | 59.23 | 25.54 | 102.94 | 4,576.15 |
| Applied to removed ratio | 1.28 | 1.68 | 1.58 | 4.89 |

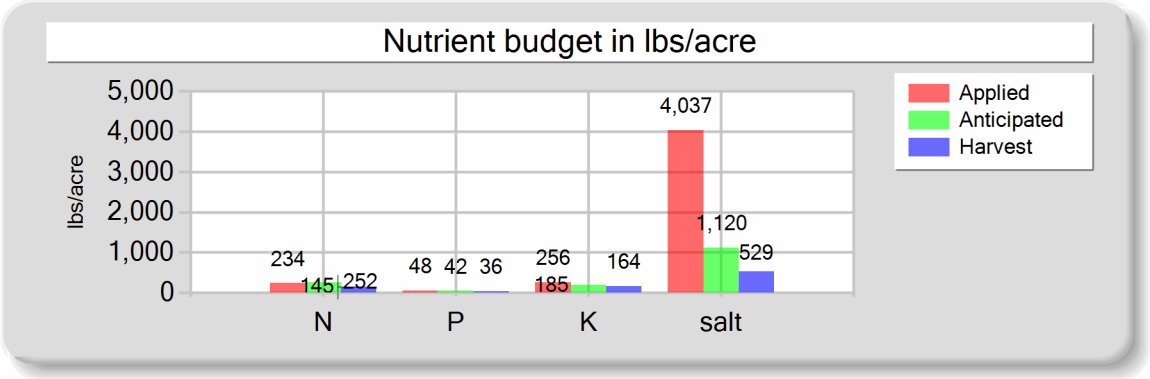
| Fresh water applied | |
|---|---|
| 20,880,000.00 | *gallons* |
| 768.94 | *acre-inches* |
| 13.26 | *inches/acre* |

| Process wastewater applied | |
|---|---|
| 0.00 | *gallons* |
| 0.00 | *acre-inches* |
| 0.00 | *inches/acre* |

| Total harvests for the crop | |
|---|---|
| 1 | *harvests* |

## Field 5 - 06/20/2023: Corn, silage

Field name: Field 5 Crop: Corn, silage Plant date: 06/20/2023




| | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Total salt (lbs/acre) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 0.00 | 0.00 | 0.00 | 0.00 |
| Dry manure | 166.94 | 47.68 | 256.07 | 0.00 |
| Process wastewater | 0.00 | 0.00 | 0.00 | 0.00 |
| Fresh water | 59.88 | 0.00 | 0.00 | 4,036.98 |
| Atmospheric deposition | 7.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 233.83 | 47.68 | 256.07 | 4,036.98 |
| Anticipated crop nutrient removal | 252.00 | 42.00 | 184.80 | 1,120.00 |
| Actual crop nutrient removal | 145.17 | 36.50 | 164.17 | 529.14 |
| Nutrient balance | 88.65 | 11.18 | 91.90 | 3,507.84 |
| Applied to removed ratio | 1.61 | 1.31 | 1.56 | 7.63 |

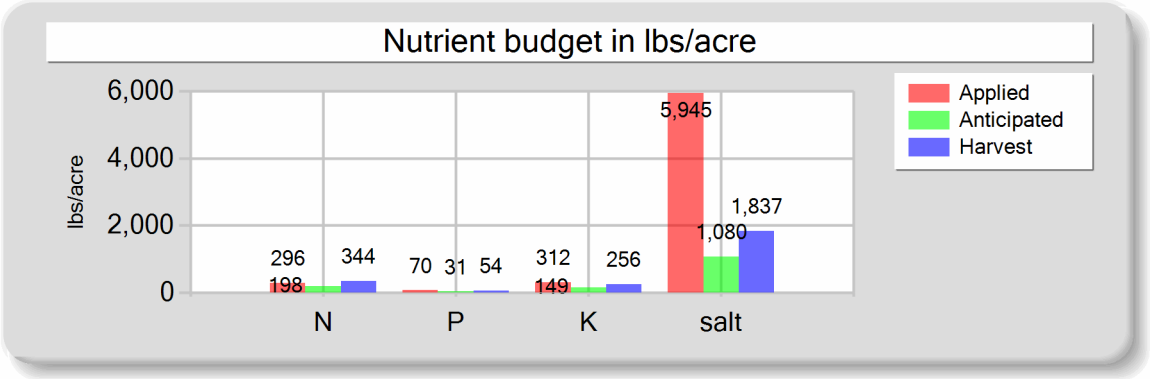
| Fresh water applied | |
|---|---|
| 56,973,000.00 | *gallons* |
| 2,098.12 | *acre-inches* |
| 36.17 | *inches/acre* |

| Process wastewater applied | |
|---|---|
| 0.00 | *gallons* |
| 0.00 | *acre-inches* |
| 0.00 | *inches/acre* |

| Total harvests for the crop | |
|---|---|
| 1 | *harvests* |

## Field 6 - 11/28/2022: Wheat, silage, soft dough

Field name: Field 6 Crop: Wheat, silage, soft dough Plant date: 11/28/2022




| | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Total salt (lbs/acre) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 0.00 | 0.00 | 0.00 | 0.00 |
| Dry manure | 272.39 | 70.31 | 311.66 | 4,814.31 |
| Process wastewater | 0.00 | 0.00 | 0.00 | 0.00 |
| Fresh water | 16.63 | 0.00 | 0.00 | 1,130.76 |
| Atmospheric deposition | 7.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 296.01 | 70.31 | 311.66 | 5,945.06 |
| Anticipated crop nutrient removal | 198.00 | 30.60 | 149.40 | 1,080.00 |
| Actual crop nutrient removal | 344.08 | 54.10 | 256.28 | 1,836.67 |
| Nutrient balance | -48.06 | 16.21 | 55.38 | 4,108.40 |
| Applied to removed ratio | 0.86 | 1.30 | 1.22 | 3.24 |

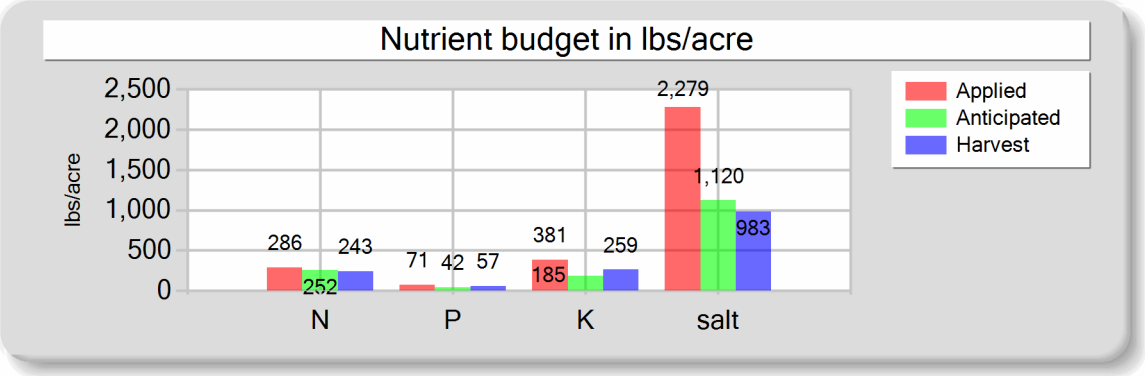
| Fresh water applied | |
|---|---|
| 7,222,500.00 | *gallons* |
| 265.98 | *acre-inches* |
| 10.23 | *inches/acre* |

| Process wastewater applied | |
|---|---|
| 0.00 | *gallons* |
| 0.00 | *acre-inches* |
| 0.00 | *inches/acre* |

| Total harvests for the crop | |
|---|---|
| 1 | *harvests* |

## Field 6 - 06/19/2023: Corn, silage

Field name: Field 6 Crop: Corn, silage Plant date: 06/19/2023




| | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Total salt (lbs/acre) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 0.00 | 0.00 | 0.00 | 0.00 |
| Dry manure | 248.27 | 70.90 | 380.82 | 0.00 |
| Process wastewater | 0.00 | 0.00 | 0.00 | 0.00 |
| Fresh water | 31.19 | 0.00 | 0.00 | 2,279.23 |
| Atmospheric deposition | 7.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 286.46 | 70.90 | 380.82 | 2,279.23 |
| Anticipated crop nutrient removal | 252.00 | 42.00 | 184.80 | 1,120.00 |
| Actual crop nutrient removal | 242.73 | 56.71 | 259.47 | 983.47 |
| Nutrient balance | 43.73 | 14.20 | 121.35 | 1,295.76 |
| Applied to removed ratio | 1.18 | 1.25 | 1.47 | 2.32 |

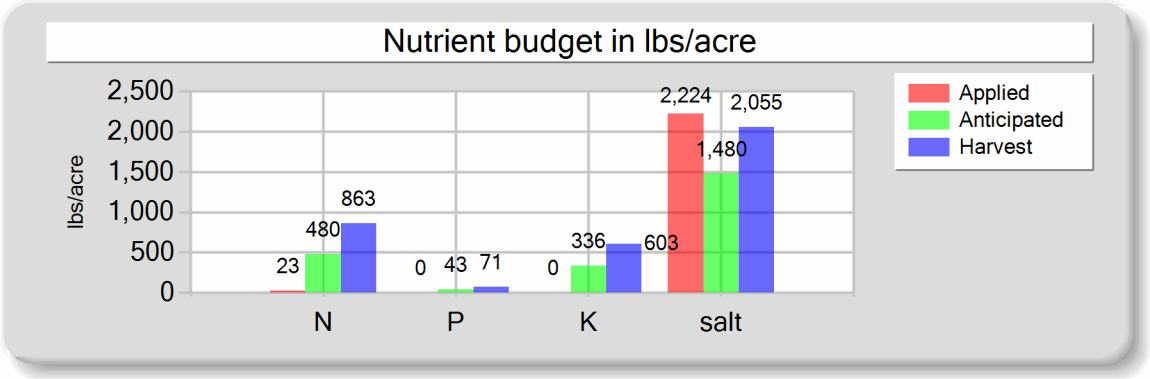
| Fresh water applied | |
|---|---|
| 15,645,000.00 | *gallons* |
| 576.15 | *acre-inches* |
| 22.16 | *inches/acre* |

| Process wastewater applied | |
|---|---|
| 0.00 | *gallons* |
| 0.00 | *acre-inches* |
| 0.00 | *inches/acre* |

| Total harvests for the crop | |
|---|---|
| 1 | *harvests* |

## Field 7 - 10/18/2020: Alfalfa, hay

Field name: Field 7 Crop: Alfalfa, hay Plant date: 10/18/2020




| | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Total salt (lbs/acre) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 0.00 | 0.00 | 0.00 | 0.00 |
| Dry manure | 0.00 | 0.00 | 0.00 | 0.00 |
| Process wastewater | 0.00 | 0.00 | 0.00 | 0.00 |
| Fresh water | 8.72 | 0.00 | 0.00 | 2,224.05 |
| Atmospheric deposition | 14.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 22.72 | 0.00 | 0.00 | 2,224.05 |
| Anticipated crop nutrient removal | 480.00 | 43.20 | 336.00 | 1,480.00 |
| Actual crop nutrient removal | 863.29 | 71.06 | 602.78 | 2,055.46 |
| Nutrient balance | -840.56 | -71.06 | -602.78 | 168.60 |
| Applied to removed ratio | 0.03 | 0.00 | 0.00 | 1.08 |

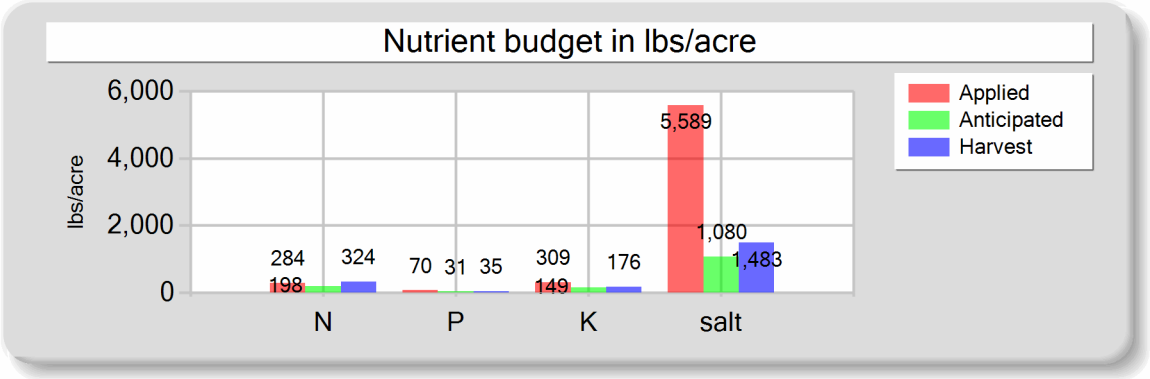
| Fresh water applied | |
|---|---|
| 68,269,500.00 | *gallons* |
| 2,514.13 | *acre-inches* |
| 34.92 | *inches/acre* |

| Process wastewater applied | |
|---|---|
| 0.00 | *gallons* |
| 0.00 | *acre-inches* |
| 0.00 | *inches/acre* |

| Total harvests for the crop | |
|---|---|
| 8 | *harvests* |

## Field 8 - 10/11/2022: Wheat, silage, soft dough

Field name: Field 8 Crop: Wheat, silage, soft dough Plant date: 10/11/2022




| | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Total salt (lbs/acre) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 0.00 | 0.00 | 0.00 | 0.00 |
| Dry manure | 270.06 | 69.71 | 309.00 | 4,773.23 |
| Process wastewater | 0.00 | 0.00 | 0.00 | 0.00 |
| Fresh water | 7.05 | 0.00 | 0.00 | 815.31 |
| Atmospheric deposition | 7.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 284.11 | 69.71 | 309.00 | 5,588.54 |
| Anticipated crop nutrient removal | 198.00 | 30.60 | 149.40 | 1,080.00 |
| Actual crop nutrient removal | 323.97 | 34.84 | 175.82 | 1,483.11 |
| Nutrient balance | -39.86 | 34.88 | 133.18 | 4,105.43 |
| Applied to removed ratio | 0.88 | 2.00 | 1.76 | 3.77 |

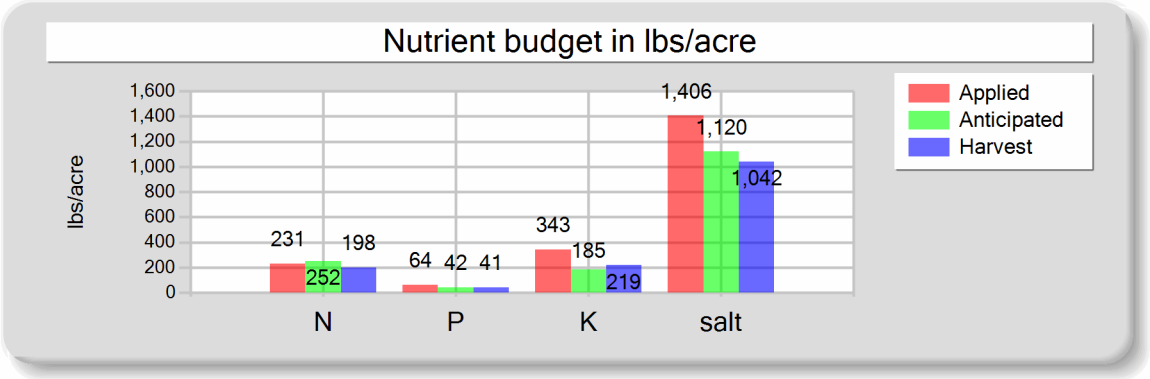
| Fresh water applied | |
|---|---|
| 34,209,000.00 | *gallons* |
| 1,259.80 | *acre-inches* |
| 16.80 | *inches/acre* |

| Process wastewater applied | |
|---|---|
| 0.00 | *gallons* |
| 0.00 | *acre-inches* |
| 0.00 | *inches/acre* |

| Total harvests for the crop | |
|---|---|
| 1 | *harvests* |

## Field 8 - 06/16/2023: Corn, silage

Field name: Field 8 Crop: Corn, silage Plant date: 06/16/2023




| | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Total salt (lbs/acre) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 0.00 | 0.00 | 0.00 | 0.00 |
| Dry manure | 223.44 | 63.81 | 342.74 | 0.00 |
| Process wastewater | 0.00 | 0.00 | 0.00 | 0.00 |
| Fresh water | 0.09 | 0.00 | 0.00 | 1,405.57 |
| Atmospheric deposition | 7.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 230.54 | 63.81 | 342.74 | 1,405.57 |
| Anticipated crop nutrient removal | 252.00 | 42.00 | 184.80 | 1,120.00 |
| Actual crop nutrient removal | 198.26 | 40.88 | 218.70 | 1,042.41 |
| Nutrient balance | 32.28 | 22.94 | 124.04 | 363.16 |
| Applied to removed ratio | 1.16 | 1.56 | 1.57 | 1.35 |

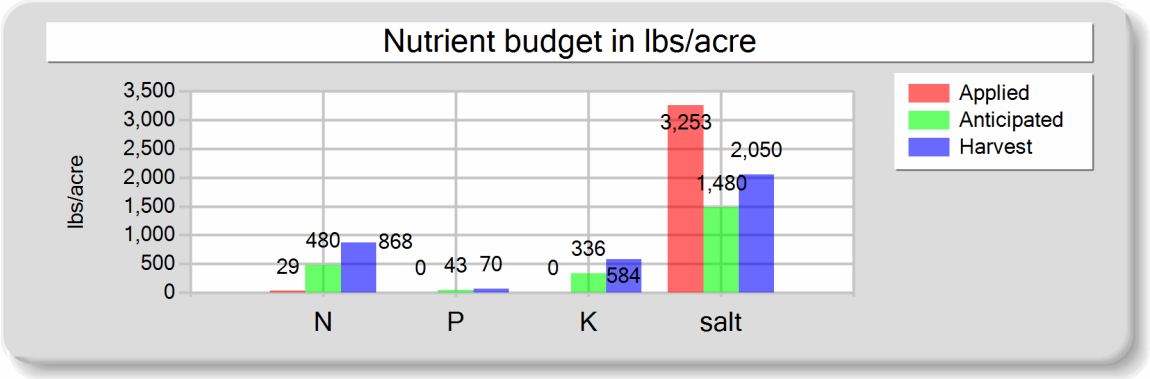
| Fresh water applied | |
|---|---|
| 55,917,000.00 | *gallons* |
| 2,059.23 | *acre-inches* |
| 27.46 | *inches/acre* |

| Process wastewater applied | |
|---|---|
| 0.00 | *gallons* |
| 0.00 | *acre-inches* |
| 0.00 | *inches/acre* |

| Total harvests for the crop | |
|---|---|
| 1 | *harvests* |

## Field 9 - 10/19/2020: Alfalfa, hay

Field name: Field 9 Crop: Alfalfa, hay Plant date: 10/19/2020




| | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Total salt (lbs/acre) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 0.00 | 0.00 | 0.00 | 0.00 |
| Dry manure | 0.00 | 0.00 | 0.00 | 0.00 |
| Process wastewater | 0.00 | 0.00 | 0.00 | 0.00 |
| Fresh water | 15.30 | 0.00 | 0.00 | 3,252.50 |
| Atmospheric deposition | 14.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 29.30 | 0.00 | 0.00 | 3,252.50 |
| Anticipated crop nutrient removal | 480.00 | 43.20 | 336.00 | 1,480.00 |
| Actual crop nutrient removal | 868.43 | 69.72 | 583.98 | 2,050.17 |
| Nutrient balance | -839.13 | -69.72 | -583.98 | 1,202.34 |
| Applied to removed ratio | 0.03 | 0.00 | 0.00 | 1.59 |

| Fresh water applied | |
|---|---|
| 56,052,000.00 | *gallons* |
| 2,064.20 | *acre-inches* |
| 51.61 | *inches/acre* |

| Process wastewater applied | |
|---|---|
| 0.00 | *gallons* |
| 0.00 | *acre-inches* |
| 0.00 | *inches/acre* |

| Total harvests for the crop | |
|---|---|
| 8 | *harvests* |

NUTRIENT ANALYSES

## A. MANURE ANALYSES

### Corral solids

Sample and source description: Corral solids

Sample date: 09/21/2022 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 45.1 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Calcium (mg/kg) | Magnesium (mg/kg) | Sodium (mg/kg) | Sulfur (mg/kg) | Chloride (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 21,500.00 | 5,550.00 | 24,600.00 | | | | | | | 38.00 |
| **DL** | 1,000.00 | 1,000.00 | 1,000.00 | | | | | | | 0.10 |

### Lagoon Manures

Sample and source description: Lagoon Manures

Sample date: 09/21/2022 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 81.3 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Calcium (mg/kg) | Magnesium (mg/kg) | Sodium (mg/kg) | Sulfur (mg/kg) | Chloride (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 23,600.00 | 2,560.00 | 5,750.00 | | | | | | | 0.00 |
| **DL** | 1,000.00 | 1,000.00 | 1,000.00 | | | | | | | 0.10 |

### Corral Solids

Sample and source description: Corral Solids

Sample date: 06/08/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 47.4 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Calcium (mg/kg) | Magnesium (mg/kg) | Sodium (mg/kg) | Sulfur (mg/kg) | Chloride (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 11,800.00 | 3,370.00 | 18,100.00 | | | | | | | 0.00 |
| **DL** | 1,000.00 | 1,000.00 | 1,000.00 | | | | | | | 0.10 |

LG-Solids

Sample and source description: LG-Solids

Sample date: 06/08/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 69.8 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Calcium (mg/kg) | Magnesium (mg/kg) | Sodium (mg/kg) | Sulfur (mg/kg) | Chloride (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 19,500.00 | 2,760.00 | 10,000.00 | | | | | | | 0.00 |
| **DL** | 1,000.00 | 1,000.00 | 1,000.00 | | | | | | | 0.10 |

Corral Solids

Sample and source description: Corral Solids

Sample date: 10/26/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 37.3 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Calcium (mg/kg) | Magnesium (mg/kg) | Sodium (mg/kg) | Sulfur (mg/kg) | Chloride (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 16,100.00 | 2,580.00 | 20,700.00 | | | | | | | 0.00 |
| **DL** | 1,000.00 | 1,000.00 | 1,000.00 | | | | | | | 0.10 |

LG-Solids

Sample and source description: LG-Solids

Sample date: 10/26/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 63.3 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Calcium (mg/kg) | Magnesium (mg/kg) | Sodium (mg/kg) | Sulfur (mg/kg) | Chloride (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 20,500.00 | 4,040.00 | 10,800.00 | | | | | | | 0.00 |
| **DL** | 1,000.00 | 1,000.00 | 1,000.00 | | | | | | | 0.10 |

**B. PROCESS WASTEWATER ANALYSES**

## LG-2-4, 2022

Sample and source description: LG-2-4, 2022

Sample date: 11/17/2022 Material type: Process wastewater Source of analysis: Lab analysis pH:

| | Kjeldahl-N (mg/L) | NH4-N (mg/L) | NH3-N (mg/L) | Nitrate-N (mg/L) | Total P (mg/L) | Total K (mg/L) | Calcium (mg/L) | Magnes. (mg/L) | Sodium (mg/L) | Bicarb. (mg/L) | Carb. (mg/L) | Sulfate (mg/L) | Chloride (mg/L) | EC (µmhos/cm) | TDS (mg/L) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 347.00 | 207.00 | 0.00 | 0.00 | 78.40 | 560.00 | | | | | | | | 4,270.00 | 2,730 |
| **DL** | 0.50 | 0.50 | 0.50 | 0.50 | 0.50 | 2.00 | | | | | | | | 1.00 | 10 |

## LG-2-1

Sample and source description: LG-2-1

Sample date: 03/16/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

| | Kjeldahl-N (mg/L) | NH4-N (mg/L) | NH3-N (mg/L) | Nitrate-N (mg/L) | Total P (mg/L) | Total K (mg/L) | Calcium (mg/L) | Magnes. (mg/L) | Sodium (mg/L) | Bicarb. (mg/L) | Carb. (mg/L) | Sulfate (mg/L) | Chloride (mg/L) | EC (µmhos/cm) | TDS (mg/L) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 350.00 | 218.00 | 0.00 | 0.00 | 50.00 | 492.00 | | | | | | | | 4,610.00 | 3,580 |
| **DL** | 0.50 | 0.50 | 0.50 | 0.50 | 0.50 | 2.00 | | | | | | | | 1.00 | 10 |

## LG-2-2

Sample and source description: LG-2-2

Sample date: 06/08/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

| | Kjeldahl-N (mg/L) | NH4-N (mg/L) | NH3-N (mg/L) | Nitrate-N (mg/L) | Total P (mg/L) | Total K (mg/L) | Calcium (mg/L) | Magnes. (mg/L) | Sodium (mg/L) | Bicarb. (mg/L) | Carb. (mg/L) | Sulfate (mg/L) | Chloride (mg/L) | EC (µmhos/cm) | TDS (mg/L) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 434.00 | 213.00 | 0.00 | 0.00 | 67.20 | 564.00 | | | | | | | | 4,320.00 | 2,760 |
| **DL** | 0.50 | 0.50 | 0.50 | 0.50 | 0.50 | 10.00 | | | | | | | | 1.00 | 10 |

## LG-2-3

Sample and source description: LG-2-3

Sample date: 08/23/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

| | Kjeldahl-N (mg/L) | NH4-N (mg/L) | NH3-N (mg/L) | Nitrate-N (mg/L) | Total P (mg/L) | Total K (mg/L) | Calcium (mg/L) | Magnes. (mg/L) | Sodium (mg/L) | Bicarb. (mg/L) | Carb. (mg/L) | Sulfate (mg/L) | Chloride (mg/L) | EC (µmhos/cm) | TDS (mg/L) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 202.00 | 70.00 | 0.00 | 0.00 | 6.59 | 108.00 | | | | | | | | 3,710.00 | 2,380 |
| **DL** | 0.50 | 0.50 | 0.50 | 0.50 | 0.50 | 2.00 | | | | | | | | 1.00 | 10 |

LG-1-4

Sample and source description: LG-1-4

Sample date: 10/26/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

| | Kjeldahl-N (mg/L) | NH4-N (mg/L) | NH3-N (mg/L) | Nitrate-N (mg/L) | Total P (mg/L) | Total K (mg/L) | Calcium (mg/L) | Magnes. (mg/L) | Sodium (mg/L) | Bicarb. (mg/L) | Carb. (mg/L) | Sulfate (mg/L) | Chloride (mg/L) | EC (µmhos/cm) | TDS (mg/L) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 342.00 | 232.00 | 0.00 | 0.00 | 71.60 | 389.00 | | | | | | | | 4,220.00 | 2,780 |
| **DL** | 0.50 | 0.50 | 0.50 | 0.50 | 0.50 | 2.00 | | | | | | | | 1.00 | 10 |

## C. FRESH WATER ANALYSES

D-1,2

Domestic Well #1,2

Sample description: Domestic Well #1,2

Sample date: 08/29/2023 Source of analysis: Lab analysis

| | Total N (mg/L) | NH4-N (mg/L) | Nitrate-N (mg/L) | Calcium (mg/L) | Magnesium (mg/L) | Sodium (mg/L) | Bicarbonate (mg/L) | Carbonate (mg/L) | Sulfate (mg/L) | Chloride (mg/L) | EC (µmhos/cm) | TDS (mg/L) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 9.98 | 0.00 | 9.98 | | | | | | | | 784.00 | 502 |
| **DL** | 0.50 | 0.50 | 0.50 | | | | | | | | 1.00 | 10 |

D-3

Domestic Well #3

Sample description: Domestic Well #3

Sample date: 08/29/2023 Source of analysis: Lab analysis

| | Total N (mg/L) | NH4-N (mg/L) | Nitrate-N (mg/L) | Calcium (mg/L) | Magnesium (mg/L) | Sodium (mg/L) | Bicarbonate (mg/L) | Carbonate (mg/L) | Sulfate (mg/L) | Chloride (mg/L) | EC (µmhos/cm) | TDS (mg/L) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 6.76 | 0.00 | 6.76 | | | | | | | | 681.00 | 436 |
| **DL** | 0.50 | 0.50 | 0.50 | | | | | | | | 1.00 | 10 |

DW-2

## DW-2

### DW-2

Sample description: DW-2

Sample date: 05/24/2023 Source of analysis: Lab analysis

| | Total N (mg/L) | NH4-N (mg/L) | Nitrate-N (mg/L) | Calcium (mg/L) | Magnesium (mg/L) | Sodium (mg/L) | Bicarbonate (mg/L) | Carbonate (mg/L) | Sulfate (mg/L) | Chloride (mg/L) | EC (µmhos/cm) | TDS (mg/L) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 1.86 | 0.00 | 1.86 | | | | | | | | 562.00 | 360 |
| **DL** | 0.50 | 0.50 | 0.50 | | | | | | | | 1.00 | 10 |

## MID Canal

### MID Canal

Sample description: MID Canal

Sample date: 06/01/2023 Source of analysis: Lab analysis

| | Total N (mg/L) | NH4-N (mg/L) | Nitrate-N (mg/L) | Calcium (mg/L) | Magnesium (mg/L) | Sodium (mg/L) | Bicarbonate (mg/L) | Carbonate (mg/L) | Sulfate (mg/L) | Chloride (mg/L) | EC (µmhos/cm) | TDS (mg/L) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 0.00 | 0.00 | 0.00 | | | | | | | | 367.00 | 235 |
| **DL** | 0.50 | 0.50 | 0.50 | | | | | | | | 1.00 | 10 |

## Well #6-1

### WELL 6-1

Sample description: WELL 6-1

Sample date: 08/23/2023 Source of analysis: Lab analysis

| | Total N (mg/L) | NH4-N (mg/L) | Nitrate-N (mg/L) | Calcium (mg/L) | Magnesium (mg/L) | Sodium (mg/L) | Bicarbonate (mg/L) | Carbonate (mg/L) | Sulfate (mg/L) | Chloride (mg/L) | EC (µmhos/cm) | TDS (mg/L) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 14.30 | 0.00 | 14.30 | | | | | | | | 1,160.00 | 739 |
| **DL** | 0.50 | 0.50 | 0.50 | | | | | | | | 1.00 | 10 |

## Well #7-1

Well #7-1

WELL 7-1

Sample description: WELL 7-1

Sample date: 08/23/2023 Source of analysis: Lab analysis

| | Total N (mg/L) | NH4-N (mg/L) | Nitrate-N (mg/L) | Calcium (mg/L) | Magnesium (mg/L) | Sodium (mg/L) | Bicarbonate (mg/L) | Carbonate (mg/L) | Sulfate (mg/L) | Chloride (mg/L) | EC (µmhos/cm) | TDS (mg/L) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 3.23 | 0.00 | 3.23 | | | | | | | | 578.00 | 370 |
| **DL** | 0.50 | 0.50 | 0.50 | | | | | | | | 1.00 | 10 |

Well #8

WELL 8

Sample description: WELL 8

Sample date: 06/01/2023 Source of analysis: Lab analysis

| | Total N (mg/L) | NH4-N (mg/L) | Nitrate-N (mg/L) | Calcium (mg/L) | Magnesium (mg/L) | Sodium (mg/L) | Bicarbonate (mg/L) | Carbonate (mg/L) | Sulfate (mg/L) | Chloride (mg/L) | EC (µmhos/cm) | TDS (mg/L) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 0.32 | 0.00 | 0.32 | | | | | | | | 10.00 | 41 |
| **DL** | 0.30 | 0.50 | 0.30 | | | | | | | | 1.00 | 10 |

## D. SOIL ANALYSES

*No soil analyses entered.*

## E. PLANT TISSUE ANALYSES

Field 1 - 10/14/2021: Alfalfa, hay

## Field 1 - 10/14/2021: Alfalfa, hay

### Alfalfa Hay 3rd cut

Sample and source description: Alfalfa Hay 3rd cut

Sample date: 06/06/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 7.5 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 38,000.00 | 3,800.00 | 37,000.00 | | 11.70 |
| **DL** | 1,000.00 | 1,000.00 | 1,000.00 | | 0.10 |

### Alfalfa Hay 5th cut

Sample and source description: Alfalfa Hay 5th cut

Sample date: 08/11/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 7.9 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 34,600.00 | 3,700.00 | 37,000.00 | | 11.70 |
| **DL** | 1,000.00 | 1,000.00 | 1,000.00 | | 0.10 |

## Field 10 - 10/11/2022: Wheat, silage, soft dough

### Winter Forage

Sample and source description: Winter Forage

Sample date: 05/23/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 56.6 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 13,500.00 | 2,110.00 | 10,500.00 | | 8.45 |
| **DL** | 1,000.00 | 1,000.00 | 1,000.00 | | 0.10 |

## Field 10 - 06/15/2023: Corn, silage

## Field 10 - 06/15/2023: Corn, silage

### Corn Silages

Sample and source description: Corn Silages

Sample date: 09/29/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 66.4 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 9,600.00 | 2,050.00 | 14,600.00 | | 6.60 |
| **DL** | 1,000.00 | 1,000.00 | 1,000.00 | | 0.10 |

## Field 11 - 10/14/2022: Wheat, silage, soft dough

### Winter Forage

Sample and source description: Winter Forage

Sample date: 04/18/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 72.0 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 20,400.00 | 5,100.00 | 57,800.00 | | 13.20 |
| **DL** | 1,000.00 | 1,000.00 | 1,000.00 | | 0.10 |

## Field 11 - 05/02/2023: Sudangrass, silage

### Sudan Silage

Sample and source description: Sudan Silage

Sample date: 08/19/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 74.6 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 18,500.00 | 3,190.00 | 29,100.00 | | 13.30 |
| **DL** | 1,000.00 | 1,000.00 | 1,000.00 | | 0.10 |

## Field 2 - 10/14/2022: Wheat, silage, soft dough

### Winter Forage

Sample and source description: Winter Forage

Sample date: 04/18/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 68.4 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 20,600.00 | 5,300.00 | 64,900.00 | | 14.70 |
| **DL** | 1,000.00 | 1,000.00 | 1,000.00 | | 0.10 |

## Field 2 - 05/12/2023: Corn, silage

### Corn Silages

Sample and source description: Corn Silages

Sample date: 08/24/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 68.2 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 9,300.00 | 1,290.00 | 17,900.00 | | 6.27 |
| **DL** | 1,000.00 | 1,000.00 | 1,000.00 | | 0.10 |

## Field 234 - 08/30/2023: Sudangrass, silage

### Sudan Silage

Sample and source description: Sudan Silage

Sample date: 10/24/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 81.6 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 18,100.00 | 3,060.00 | 30,400.00 | | 24.60 |
| **DL** | 1,000.00 | 1,000.00 | 1,000.00 | | 0.10 |

## Field 3 - 10/14/2022: Wheat, silage, soft dough

### Winter Forage

Sample and source description: Winter Forage

Sample date: 04/18/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 68.6 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 20,400.00 | 4,900.00 | 60,700.00 | | 14.60 |
| **DL** | 1,000.00 | 1,000.00 | 1,000.00 | | 0.10 |

## Field 3 - 05/12/2023: Corn, silage

### Corn Silages

Sample and source description: Corn Silages

Sample date: 08/24/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 69.2 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 10,300.00 | 2,810.00 | 14,400.00 | | 6.80 |
| **DL** | 1,000.00 | 1,000.00 | 1,000.00 | | 0.10 |

## Field 4 - 10/14/2022: Wheat, silage, soft dough

### Winter Forage

Sample and source description: Winter Forage

Sample date: 04/18/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 70.4 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 18,300.00 | 5,300.00 | 64,100.00 | | 13.70 |
| **DL** | 1,000.00 | 1,000.00 | 1,000.00 | | 0.10 |

## Field 4 - 05/12/2023: Corn, silage

### Corn Silages

Sample and source description: Corn Silages

Sample date: 08/25/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 69.0 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 9,600.00 | 2,890.00 | 14,600.00 | | 6.67 |
| **DL** | 1,000.00 | 1,000.00 | 1,000.00 | | 0.10 |

## Field 5 - 11/28/2022: Wheat, silage, soft dough

### Winter Forage

Sample and source description: Winter Forage

Sample date: 05/22/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 58.0 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 13,500.00 | 2,380.00 | 11,200.00 | | 7.47 |
| **DL** | 1,000.00 | 1,000.00 | 1,000.00 | | 0.10 |

## Field 5 - 06/20/2023: Corn, silage

### Corn Silages

Sample and source description: Corn Silages

Sample date: 09/28/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 72.6 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 10,700.00 | 2,690.00 | 12,100.00 | | 3.90 |
| **DL** | 1,000.00 | 1,000.00 | 1,000.00 | | 0.10 |

## Field 6 - 11/28/2022: Wheat, silage, soft dough

### Winter Forage

Sample and source description: Winter Forage

Sample date: 05/22/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 58.2 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 14,500.00 | 2,280.00 | 10,800.00 | | 7.74 |
| **DL** | 1,000.00 | 1,000.00 | 1,000.00 | | 0.10 |

## Field 6 - 06/19/2023: Corn, silage

### Corn Silages

Sample and source description: Corn Silages

Sample date: 09/28/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 70.4 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 11,600.00 | 2,710.00 | 12,400.00 | | 4.70 |
| **DL** | 1,000.00 | 1,000.00 | 1,000.00 | | 0.10 |

## Field 7 - 10/18/2020: Alfalfa, hay

### Alfalfa Haylage

Sample and source description: Alfalfa Haylage

Sample date: 04/19/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 60.4 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 37,200.00 | 3,000.00 | 47,000.00 | | 13.10 |
| **DL** | 1,000.00 | 1,000.00 | 1,000.00 | | 0.10 |

## Field 7 - 10/18/2020: Alfalfa, hay

### Alfalfa Hay 2nd cut

Sample and source description: Alfalfa Hay 2nd cut

Sample date: 06/05/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 7.8 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 45,700.00 | 3,400.00 | 20,000.00 | | 9.08 |
| **DL** | 1,000.00 | 1,000.00 | 1,000.00 | | 0.10 |

### Alfalfa Hay 5th cut

Sample and source description: Alfalfa Hay 5th cut

Sample date: 08/11/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 7.8 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 42,900.00 | 3,700.00 | 25,000.00 | | 8.98 |
| **DL** | 1,000.00 | 1,000.00 | 1,000.00 | | 0.10 |

## Field 8 - 10/11/2022: Wheat, silage, soft dough

### Winter Forage

Sample and source description: Winter Forage

Sample date: 05/23/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 55.6 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 19,900.00 | 2,140.00 | 10,800.00 | | 9.11 |
| **DL** | 1,000.00 | 1,000.00 | 1,000.00 | | 0.10 |

## Field 8 - 06/16/2023: Corn, silage

## Field 8 - 06/16/2023: Corn, silage

### Corn Silages

Sample and source description: Corn Silages

Sample date: 09/29/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 64.2 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 9,700.00 | 2,000.00 | 10,700.00 | | 5.10 |
| **DL** | 1,000.00 | 1,000.00 | 1,000.00 | | 0.10 |

## Field 9 - 10/19/2020: Alfalfa, hay

### Alfalfa Haylage

Sample and source description: Alfalfa Haylage

Sample date: 04/19/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 60.4 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 37,200.00 | 3,000.00 | 47,000.00 | | 13.10 |
| **DL** | 1,000.00 | 1,000.00 | 1,000.00 | | 0.10 |

### Alfalfa Hay 2nd cut

Sample and source description: Alfalfa Hay 2nd cut

Sample date: 06/05/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 7.8 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 45,700.00 | 3,400.00 | 20,000.00 | | 9.08 |
| **DL** | 1,000.00 | 1,000.00 | 1,000.00 | | 0.10 |

Field 9 - 10/19/2020: Alfalfa, hay

Alfalfa Hay 5th cut

Sample and source description: Alfalfa Hay 5th cut

Sample date: 08/11/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 7.8 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 42,900.00 | 3,700.00 | 25,000.00 | | 8.98 |
| **DL** | 1,000.00 | 1,000.00 | 1,000.00 | | 0.10 |

## F. SUBSURFACE (TILE) DRAINAGE ANALYSES

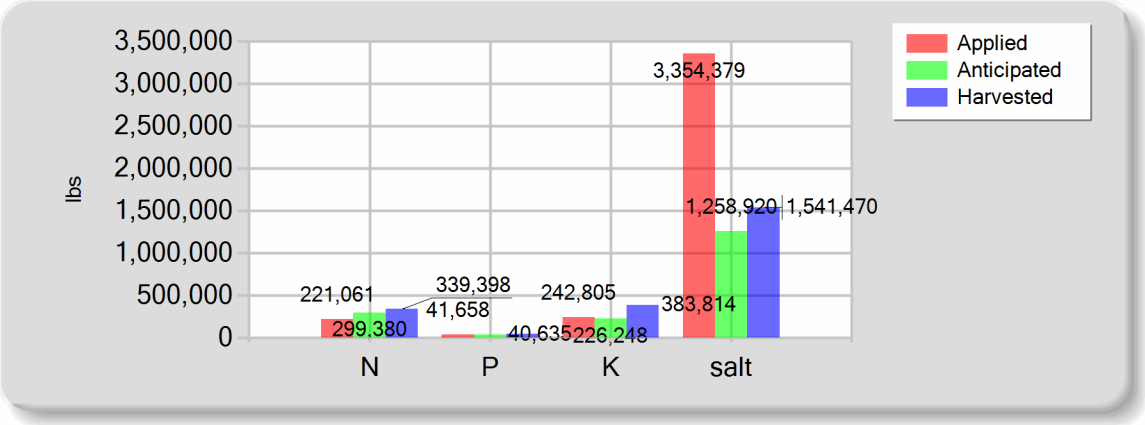
*No subsurface (tile) drainage analyses entered.*

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

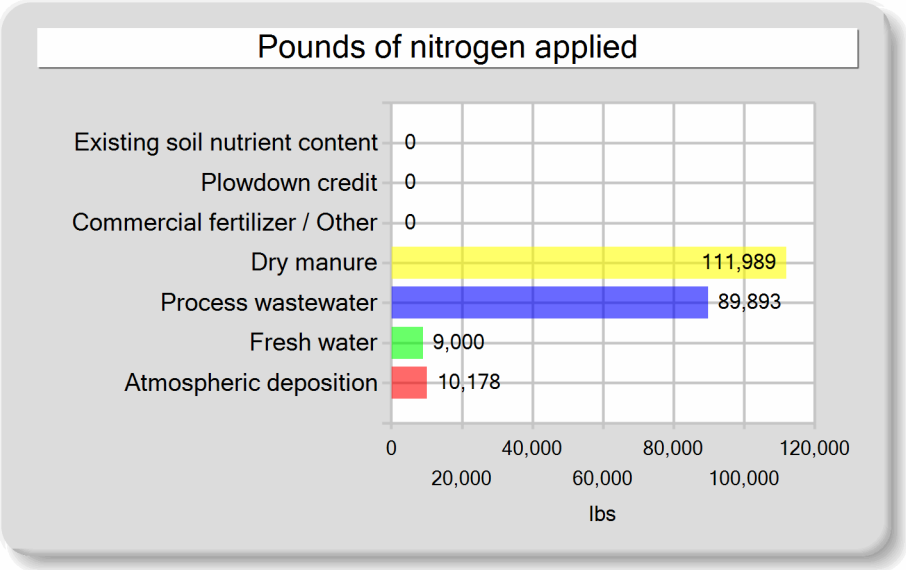
## A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

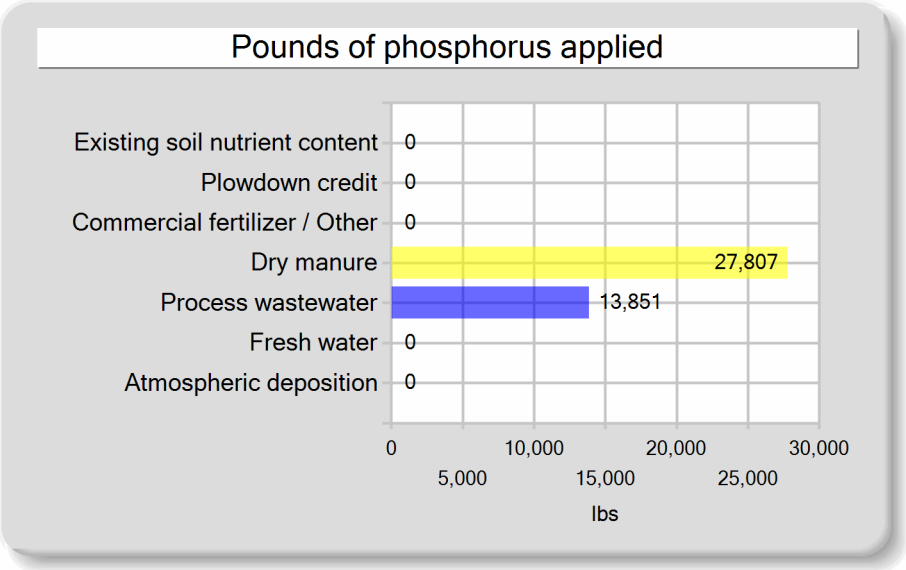
| | Total N (lbs) | Total P (lbs) | Total K (lbs) | Total salt (lbs) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 0.00 | 0.00 | 0.00 | 0.00 |
| Dry manure | 111,989.47 | 27,806.97 | 130,099.90 | 1,094,003.28 |
| Process wastewater | 89,892.87 | 13,850.88 | 112,705.57 | 748,208.03 |
| Fresh water | 9,000.15 | 0.00 | 0.00 | 1,512,167.54 |
| Atmospheric deposition | 10,178.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 221,060.50 | 41,657.85 | 242,805.46 | 3,354,378.84 |
| Anticipated crop nutrient removal | 299,380.00 | 40,634.80 | 226,248.00 | 1,258,920.00 |
| Actual crop nutrient removal | 339,397.84 | 47,581.60 | 383,813.80 | 1,541,469.88 |
| Nutrient balance | -118,337.33 | -5,923.76 | -141,008.33 | 1,812,908.97 |
| Applied to removed ratio | 0.65 | 0.88 | 0.63 | 2.18 |

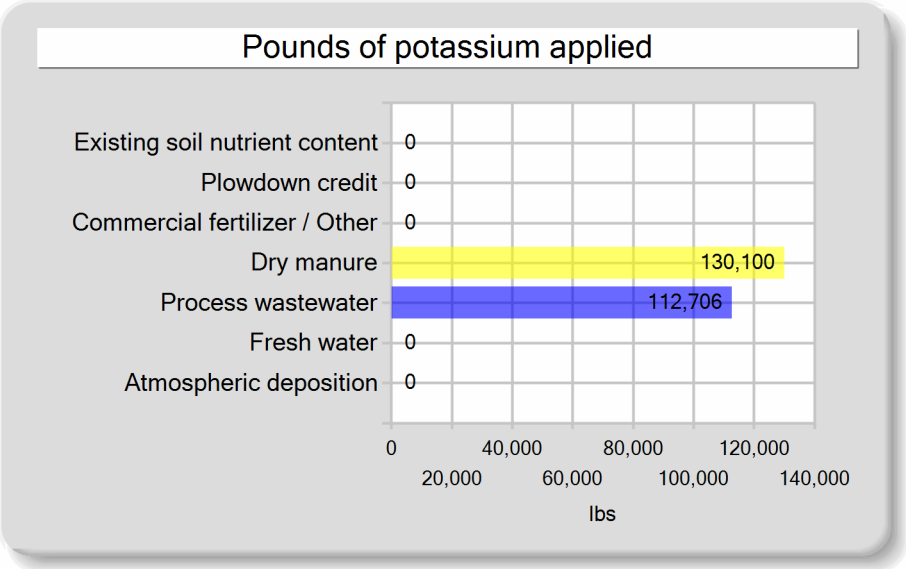
## B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

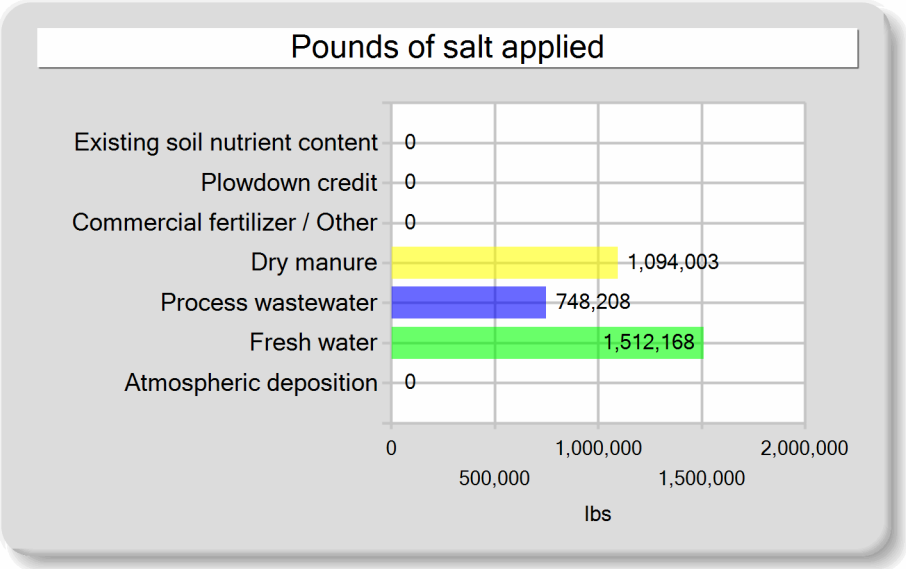


## C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE



## EXCEPTION REPORTING

### A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

*No manure or process wastewater discharges occurred during the reporting period.*

### B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

*No stormwater discharges occurred during the reporting period.*

### C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

*No land application area to surface water discharges occurred during the reporting period.*

## NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

### A. NUTRIENT MANAGEMENT PLAN STATEMENTS

| | |
|---|---|
| Was the facility's NMP updated in the reporting period? | Yes |
| Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order? | Yes |
| Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order? | Yes |

### B. EXPORT AGREEMENT STATEMENT

| | |
|---|---|
| Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period? | No |

ADDITIONAL NOTES

**A. NOTES**

CARLOS LOURENCO DAIRY:

Field 5 - Corn silage; ratio is somewhat higher because the crop removal shows as more than 100 lbs less than expected. This is because the plant tissue is much less than normal.

Much of the manures are being held as bedding.

CERTIFICATION

**A. OWNER AND/OR OPERATOR CERTIFICATION**

*I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.*

| | |
|---|---|
| Carlos Lourenco | |
| SIGNATURE OF OWNER OF FACILITY | SIGNATURE OF OPERATOR OF FACILITY |
| Carlos Lourenco | SAME AS OWNER |
| PRINT OR TYPE NAME | PRINT OR TYPE NAME |
| 06-28-2024 | |
| DATE | DATE |

ATTACHMENTS

## A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

| Annual Dairy Facility Assessment |
| --- |
| Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A ) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed . |

| Manure/Process Wastewater Tracking Manifests |
| --- |
| Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler. |

| Corrective Actions Documents |
| --- |
| Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction. |

| Groundwater Monitoring |
| --- |
| Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13. |

| Storm Water Monitoring |
| --- |
| Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14. |

**Denele Analytical, Inc.**

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

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40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

# Compliance Analysis Report

**Customer**
Carlos Lourenco Dairy
6557 Oak Ave.
Merced, CA 95341

**Grower**

**Consultant:**
Heartland Consulting
P.O. Box 3679
Merced, CA 95344

PURCHASE ORDER:
RECEIVED DATE: 5/24/23 4:07 PM
SUBMITTED BY: Dan Roos
ANALYZED DATE: 5/26/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying @ 24.6c

| Sample ID | Lab ID | Analyte | Method Ref | Result | Units |
|---|---|---|---|---|---|
| DW-2 | T3144039E | Electrical Conductivity (EC) | EPA 120.1 and | 0.562 | mmhos/cm |
| | | Nitrate Nitrogen (NO3-N) | EPA 353.2 | 1.86 | mg/L |
| | | Ammonia Nitrogen (NH3-N) | SM 4500-NH3 D,C | < 0.500 | mg/L |
| | | Soluble Salts (SALT-SOL) | Calculation | 360 | mg/L |

| Laboratory Quality Control | LCS % | MS % | MSD % | RPD % |
|---|---|---|---|---|
| Ammonia Nitrogen (NH3-N) | 99.2 | 91.0 (T3144039E) | 95.2 (T3144039E) | 4.50 |
| Electrical Conductivity (EC) | 99.4 | | | 0.000 |
| Nitrate Nitrogen (NO3-N) | 101 | 97.0 (T3143007A) | 98.6 (T3143007A) | 1.70 |

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

# DENELE ANALYTICAL, INC.

865 S. KILROY
TURLOCK, CA 95380
PH# 209-634-9055 FAX# 209-634-9057

## DAIRY COMPLIANCE

Client: CARLOS LOURENCO DAIRY
Mailing address: 6557 Oak Avenue, Merced CA 95341
(not facility address)
Phone#: 209-495-8654

Authorized Copy Released To:
hlcdan@yahoo.com

Sampled By: Daniel Roos
Project ID:

### ANALYSIS TO BE COMPLETED:

**IRRIGATION/GROUND WATER**
H1 EC*, NO3-N, NH4-N*
H6 EC, TN, TDS
H7 Combination of H1 & H6
H14 EC, NO3-N, NH4-N, Ca, Mg, Na, Bicarb Carb, SO4, Cl, TDS

**PROCESS WASTE WATER**
H3 EC*, NH4-N, TKN, TP, K, TDS
H4 EC*, NO3-N, NH4-N, TKN, TP, K, TDS
H5 Ca, Mg, Na, Bicarb, Carb, SO4, Cl
H8 Combination of H3 &H5
H9 Combination of H4 &H5

**TILE DRAIN**
H2 EC*, NO3-N, TP, NH4-N*, TDS

** Required Field Measurement*
___ (Check here if performed)

**DISCHARGE WATER**
E1 EC*, Temp*, pH*, NO3-N, NH4-N, TKN, TP, K, TDS, Dissolved Oxygen*, BOD 5, TSS, Total & Fecal Coliforms
E2 EC,* Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Dissolved Oxygen*, Total & Fecal Coliforms
E3 EC*, Temp*, pH*, NO3-N, NH4-N*, TP, Total & Fecal Coliforms
E4 EC*, Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Total & Fecal Coliform

**MANURE**
M1 % Moisture
M3 TN TP, K, % Moisture
M4 Ca, Mg, Na, Bicarb, Carb, SO4, Cl, Ash
M7 Combination of M3 & M4

**PLANT TISSUE**
F18 % Moisture, Ash, TN, P, K, DM and as received
F18+ % Moisture, Ash, TN, P, K, DM and as received (Package also includes NIR DCAD)
F19+ % Moisture, TN (Package includes complete plant tissue analysis)

**SOIL**
K3 Soluble Phosphorus
K4 1ft NO3, Organic Matter
K5 Pre-Plant (2ft NO3), Fall (1-2ft NO3), Fall (2-3ft NO3)
K6 0-1ft EC, NO3, Soluble P, K, Organic Matter
K7 Combination of K3 & K4
K8 Combination of K3 & K5

| | Date Collected | Time Collected | Sample ID | Description | Temp | Well Purge | Analysis: H1 | H3 | M3 | F18 | OTHER |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 5-22-23 | 1000 | #5 - WF | WINTER FORAGES | ~~24.6~~ | | | | | X | |
| 2 | " | 1400 | #6 - WF | " | " | | | | | X | |
| 3 | 5-23-23 | 1100 | #8 - WF | " | " | | | | | X | |
| 4 | " | 1240 | #10 - WF | " | " | | | | | X | |
| 5 | 5-24-23 | 1500 | DW-2 | | 24.6 | | X | | | | |
| 6 | | | | | | | | | | | |
| 7 | | | | | | | | | | | |
| 8 | | | | | | | | | | | |
| 9 | | | | | | | | | | | |
| 10 | | | | | | | | | | | |

### CHAIN OF CUSTODY RECORDING

| | Signature | Company | Received Date & Time | Relinquished Date & Time | (LABORATORY USE ONLY) Logged In By: | LABORATORY NUMBER |
|---|---|---|---|---|---|---|
| 1st | | HeartLand Consulting | 5-24-23 1607 | | | |
| 2nd | | DENELE ANALYTICAL, INC. | 5-24-23 4:07 | | | T3144039 |

Required to be submitted to any agency? (YES / NO)

**Denele Analytical, Inc.**

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www.denelelabs.com

# Compliance Analysis Report

**Customer**
Carlos Lourenco Dairy
6557 Oak Ave.
Merced, CA 95341

**Grower**

**Consultant:**
Heartland Consulting
P.O. Box 3679
Merced, CA 95344

PURCHASE ORDER:
RECEIVED DATE: 6/1/23 3:10 PM
SUBMITTED BY: Dan
ANALYZED DATE: 6/2/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 18.2c

| Sample ID | Lab ID | Analyte | Method Ref | Result | Units |
|---|---|---|---|---|---|
| Well #8 | T3152032A | Electrical Conductivity (EC) | EPA 120.1 and | < 0.100 | mmhos/cm |
| | | Nitrate Nitrogen (NO3-N) | EPA 353.2 | 0.318 | mg/L |
| | | Ammonia Nitrogen (NH3-N) | SM 4500-NH3 D,C | < 0.500 | mg/L |
| | | Soluble Salts (SALT-SOL) | Calculation | 40.5 | mg/L |

| Laboratory Quality Control | LCS % | MS % | MSD % | RPD % |
|---|---|---|---|---|
| Ammonia Nitrogen (NH3-N) | 94.7 | 93.9 (T3152032A) | 91.6 (T3152032A) | 2.45 |
| Electrical Conductivity (EC) | 96.8 | | | 4.36 |
| Nitrate Nitrogen (NO3-N) | 107 | 101 (T3152032A) | 95.3 (T3152032A) | 5.95 |

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ELAP Certificate No.: 2714

**Denele Analytical, Inc.**

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

www.denelelabs.com

# Compliance Analysis Report

**Customer**
Carlos Lourenco Dairy
6557 Oak Ave.
Merced, CA 95341

**Grower**

**Consultant:**
Heartland Consulting
P.O. Box 3679
Merced, CA 95344

PURCHASE ORDER:
RECEIVED DATE: 6/1/23 3:10 PM
SUBMITTED BY: Dan
ANALYZED DATE: 6/2/23

Irrigation/ Ground Water (H1)
SOURCE: Irrigation Water

NOTES: Temperature was above qualifying at 16.8c

| Sample ID | Lab ID | Analyte | Method Ref | Result | Units |
|---|---|---|---|---|---|
| MID Canal | T3152032B | Electrical Conductivity (EC) | EPA 120.1 and | 0.367 | mmhos/cm |
| | | Nitrate Nitrogen (NO3-N) | EPA 353.2 | < 0.140 | mg/L |
| | | Ammonia Nitrogen (NH3-N) | SM 4500-NH3 D,C | < 0.500 | mg/L |
| | | Soluble Salts (SALT-SOL) | Calculation | 235 | mg/L |

| Laboratory Quality Control | LCS % | MS % | MSD % | RPD % |
|---|---|---|---|---|
| Ammonia Nitrogen (NH3-N) | 94.7 | 93.9 (T3152032A) | 91.6 (T3152032A) | 2.45 |
| Electrical Conductivity (EC) | 96.8 | | | 4.36 |
| Nitrate Nitrogen (NO3-N) | 107 | 101 (T3152032A) | 95.3 (T3152032A) | 5.95 |

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ELAP Certificate No.: 2714

# DENELE ANALYTICAL, INC.

865 S. KILROY
TURLOCK, CA 95380
PH# 209-634-9055 FAX# 209-634-9057

## DAIRY COMPLIANCE

Client: CARLOS LOURENCO DAIRY
Mailing address: 6557 Oak Avenue, Merced CA 95341
(not facility address)
Phone#: 209-495-8654

Authorized Copy Released To:
hlcdan@yahoo.com

Sampled By: Daniel Roos
Project ID:

## ANALYSIS TO BE COMPLETED:

**IRRIGATION/GROUND WATER**
H1 EC*, NO3-N, NH4-N*
H6 EC, TN, TDS
H7 Combination of H1 & H6
H14 EC, NO3-N, NH4-N, Ca, Mg, Na, Bicarb Carb, SO4, Cl, TDS

**PROCESS WASTE WATER**
H3 EC*, NH4-N, TKN, TP, K, TDS
H4 EC*, NO3-N, NH4-N, TKN, TP, K, TDS
H5 Ca, Mg, Na, Bicarb, Carb, SO4, Cl
H8 Combination of H3 &H5
H9 Combination of H4 &H5

**TILE DRAIN**
H2 EC*, NO3-N, TP, NH4-N*, TDS

** Required Field Measurement*

___ (Check here if performed)

**DISCHARGE WATER**
E1 EC*, Temp*, pH*, NO3-N, NH4-N, TKN, TP, K, TDS, Dissolved Oxygen*, BOD 5, TSS, Total & Fecal Coliforms
E2 EC,* Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Dissolved Oxygen*, Total & Fecal Coliforms
E3 EC*, Temp*, pH*, NO3-N, NH4-N*, TP, Total & Fecal Coliforms
E4 EC*, Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Total & Fecal Coliform

**MANURE**
M1 % Moisture
M3 TN TP, K, % Moisture
M4 Ca, Mg, Na, Bicarb, Carb, SO4, Cl, Ash
M7 Combination of M3 & M4

**PLANT TISSUE**
F18 % Moisture, Ash, TN, P, K, DM and as received
F18+ % Moisture, Ash, TN, P, K, DM and as received (Package also includes NIR DCAD)
F19+ % Moisture, TN (Package includes complete plant tissue analysis)

**SOIL**
K3 Soluble Phosphorus
K4 1ft NO3, Organic Matter
K5 Pre-Plant (2ft NO3), Fall (1-2ft NO3), Fall (2-3ft NO3)
K6 0-1ft EC, NO3, Soluble P, K, Organic Matter
K7 Combination of K3 & K4
K8 Combination of K3 & K5

| | Date Collected | Time Collected | Sample ID | Description | Temp | Well Purge | H1 | H3 | M3 | F18 | OTHER |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 6-1-23 | 1310 | WELL #8 | | 18.2° | | X | | | | |
| 2 | 6-1-23 | 1320 | MID CANAL | | 16.8° | | X | | | | |
| 3 | | | | | | | | | | | |
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Analysis

**CHAIN OF CUSTODY RECORDING**

| | Signature | Company | Received Date & Time | Relinquished Date & Time | Logged In By: | LABORATORY NUMBER |
|---|---|---|---|---|---|---|
| 1st | [signature] | HeartLand Consulting | 6-1-23 1510 | | | |
| 2nd | [signature] | DENELE ANALYTICAL, INC. | 6-1-23 3:18 | | | T3152032 |

*(LABORATORY USE ONLY)*

Required to be submitted to any agency? (YES / NO)

**Denele Analytical, Inc.**

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

40 N. East Street, Woodland, CA 95776
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www.denelelabs.com

# Compliance Analysis Report

**Customer**
Carlos Lourenco Dairy
6557 Oak Ave.
Merced, CA 95341

**Grower**

**Consultant:**
Heartland Consulting
P.O. Box 3679
Merced, CA 95344

PURCHASE ORDER:
RECEIVED DATE: 8/23/23 3:18 PM
SUBMITTED BY: Daniel Roos
ANALYZED DATE: 8/25/23

Irrigation/ Ground Water (H1)
SOURCE: Irrigation Water

NOTES: Temperature was above qualifying at 23.2C

| Sample ID | Lab ID | Analyte | Method Ref | Result | Units |
|---|---|---|---|---|---|
| # 6-1 | T3235035C | Electrical Conductivity (EC) | EPA 120.1 and | 1.16 | mmhos/cm |
| | | Nitrate Nitrogen (NO3-N) | EPA 353.2 | 14.3 | mg/L |
| | | Ammonia Nitrogen (NH3-N) | SM 4500-NH3 D,C | < 0.500 | mg/L |
| | | Soluble Salts (SALT-SOL) | Calculation | 739 | mg/L |

| Laboratory Quality Control | LCS % | MS % | MSD % | RPD % |
|---|---|---|---|---|
| Ammonia Nitrogen (NH3-N) | 101 | 95.1 (T3235024B) | 95.9 (T3235024B) | 0.815 |
| Electrical Conductivity (EC) | 97.1 | | | 4.81 |
| Nitrate Nitrogen (NO3-N) | 104 | | | 7.24 |

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# Compliance Analysis Report

**Customer**
Carlos Lourenco Dairy
6557 Oak Ave.
Merced, CA 95341

**Grower**

**Consultant:**
Heartland Consulting
P.O. Box 3679
Merced, CA 95344

PURCHASE ORDER:
RECEIVED DATE: 8/23/23 3:18 PM
SUBMITTED BY: Daniel Roos
ANALYZED DATE: 8/25/23

Irrigation/ Ground Water (H1)
SOURCE: Irrigation Water

NOTES: Temperature was above qualifying at 22.8C

| Sample ID | Lab ID | Analyte | Method Ref | Result | Units |
|---|---|---|---|---|---|
| # 7-1 | T3235035D | Electrical Conductivity (EC) | EPA 120.1 and | 0.578 | mmhos/cm |
| | | Nitrate Nitrogen (NO3-N) | EPA 353.2 | 3.23 | mg/L |
| | | Ammonia Nitrogen (NH3-N) | SM 4500-NH3 D,C | < 0.500 | mg/L |
| | | Soluble Salts (SALT-SOL) | Calculation | 370 | mg/L |

| Laboratory Quality Control | LCS % | MS % | MSD % | RPD % |
|---|---|---|---|---|
| Ammonia Nitrogen (NH3-N) | 101 | 95.1 (T3235024B) | 95.9 (T3235024B) | 0.815 |
| Electrical Conductivity (EC) | 97.1 | | | 4.81 |
| Nitrate Nitrogen (NO3-N) | 104 | | | 7.24 |

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# DENELE ANALYTICAL, INC.

865 S. KILROY
TURLOCK, CA 95380

PH# 209-634-9055 FAX# 209-634-9057

## DAIRY COMPLIANCE

Client: CARLOS LOURENCO DAIRY
Mailing address: 6557 Oak Avenue, Merced CA 95341
(not facility address)
Phone#: 209-495-8654

Authorized Copy Released To:
hlcdan@yahoo.com

Sampled By: Daniel Roos
Project ID:

### ANALYSIS TO BE COMPLETED:

**IRRIGATION/GROUND WATER**
H1 EC*, NO3-N, NH4-N*
H6 EC, TN, TDS
H7 Combination of H1 & H6
H14 EC, NO3-N, NH4-N, Ca, Mg, Na, Bicarb Carb, SO4, Cl, TDS

**PROCESS WASTE WATER**
H3 EC*, NH4-N, TKN, TP, K, TDS
H4 EC*, NO3-N, NH4-N, TKN, TP, K, TDS
H5 Ca, Mg, Na, Bicarb, Carb, SO4, Cl
H8 Combination of H3 &H5
H9 Combination of H4 &H5

**TILE DRAIN**
H2 EC*, NO3-N, TP, NH4-N*, TDS

** Required Field Measurement*

___ (Check here if performed)

**DISCHARGE WATER**
E1 EC*, Temp*, pH*, NO3-N, NH4-N, TKN, TP, K, TDS, Dissolved Oxygen*, BOD 5, TSS, Total & Fecal Coliforms
E2 EC,* Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Dissolved Oxygen*, Total & Fecal Coliforms
E3 EC*, Temp*, pH*, NO3-N, NH4-N*, TP, Total & Fecal Coliforms
E4 EC*, Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Total & Fecal Coliform

**MANURE**
M1 % Moisture
M3 TN TP, K, % Moisture
M4 Ca, Mg, Na, Bicarb, Carb, SO4, Cl, Ash
M7 Combination of M3 & M4

**PLANT TISSUE**
F18 % Moisture, Ash, TN, P, K, DM and as received
F18+ % Moisture, Ash, TN, P, K, DM and as received (Package also includes NIR DCAD)
F19+ % Moisture, TN (Package includes complete plant tissue analysis)

**SOIL**
K3 Soluble Phosphorus
K4 1ft NO3, Organic Matter
K5 Pre-Plant (2ft NO3), Fall (1-2ft NO3), Fall (2-3ft NO3)
K6 0-1ft EC, NO3, Soluble P, K, Organic Matter
K7 Combination of K3 & K4
K8 Combination of K3 & K5

| | Date Collected | Time Collected | Sample ID | Description | Temp | Well Purge | Analysis: H1 | H3 | M3 | F18 | OTHER |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 8-19-23 | 1200 | #11-SUDAN | | | | | | | X | |
| 2 | | | | | | | | | | | |
| 3 | 8-23-23 | 0900 | 6-2-3 | | 23.4C | | | X | | | |
| 4 | 8-23-23 | 0925 | # 6-1 | | 23.2C | | X | | | | |
| 5 | 8-23-23 | 0930 | # 7-1 | | 22.8C | | X | | | | |
| 6 | | | | | | | | | | | |
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| 10 | | | | | | | | | | | |

**CHAIN OF CUSTODY RECORDING**

| | Signature | Company | Received Date & Time | Relinquished Date & Time | (LABORATORY USE ONLY) Logged In By: | LABORATORY NUMBER |
|---|---|---|---|---|---|---|
| 1st | | HeartLand Consulting | | 8-23-23 1430 | | |
| 2nd | | DENELE ANALYTICAL, INC. | 8/23/23 1430 | | | T3235035 |

Required to be submitted to any agency? (YES / NO)

**Denele Analytical, Inc.**

865 South Kilroy Avenue Turlock, CA 95380
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# Compliance Analysis Report

**Customer**
Carlos Lourenco Dairy
6557 Oak Ave.
Merced, CA 95341

**Grower**

**Consultant:**
Heartland Consulting
P.O. Box 3679
Merced, CA 95344

PURCHASE ORDER:
RECEIVED DATE: 8/29/23 1:30 PM
SUBMITTED BY: Dan
ANALYZED DATE: 8/30/23

Irrigation/ Ground Water (H1)
SOURCE: Irrigation Water

NOTES: Temperature was above qualifying at 18.0c

| Sample ID | Lab ID | Analyte | Method Ref | Result | Units |
|---|---|---|---|---|---|
| D-#3 | T3241016D | Electrical Conductivity (EC) | EPA 120.1 and | 0.681 | mmhos/cm |
| | | Nitrate Nitrogen (NO3-N) | EPA 353.2 | 6.76 | mg/L |
| | | Ammonia Nitrogen (NH3-N) | SM 4500-NH3 D,C | < 0.500 | mg/L |
| | | Soluble Salts (SALT-SOL) | Calculation | 436 | mg/L |

| Laboratory Quality Control | LCS % | MS % | MSD % | RPD % |
|---|---|---|---|---|
| Ammonia Nitrogen (NH3-N) | 98.4 | 94.0 (T3241014A) | 92.2 (T3241014A) | 1.93 |
| Electrical Conductivity (EC) | 99.6 | | | 0.636 |
| Nitrate Nitrogen (NO3-N) | 99.1 | 102 (T3240032B) | 89.6 (T3240032B) | 12.6 |

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# Compliance Analysis Report

**Customer**
Carlos Lourenco Dairy
6557 Oak Ave.
Merced, CA 95341

**Grower**

**Consultant:**
Heartland Consulting
P.O. Box 3679
Merced, CA 95344

PURCHASE ORDER:
RECEIVED DATE: 8/29/23 1:30 PM
SUBMITTED BY: Dan
ANALYZED DATE: 8/30/23

Irrigation/ Ground Water (H1)
SOURCE: Irrigation Water

NOTES: Temperature was above qualifying at 17.6c

| Sample ID | Lab ID | Analyte | Method Ref | Result | Units |
|---|---|---|---|---|---|
| D-1,2 | T3241016E | Electrical Conductivity (EC) | EPA 120.1 and | 0.784 | mmhos/cm |
| | | Nitrate Nitrogen (NO3-N) | EPA 353.2 | 9.98 | mg/L |
| | | Ammonia Nitrogen (NH3-N) | SM 4500-NH3 D,C | < 0.500 | mg/L |
| | | Soluble Salts (SALT-SOL) | Calculation | 502 | mg/L |

| Laboratory Quality Control | LCS % | MS % | MSD % | RPD % |
|---|---|---|---|---|
| Ammonia Nitrogen (NH3-N) | 98.4 | 94.0 (T3241014A) | 92.2 (T3241014A) | 1.93 |
| Electrical Conductivity (EC) | 99.6 | | | 0.636 |
| Nitrate Nitrogen (NO3-N) | 98.5 | 99.5 (T3241011A) | 95.9 (T3241011A) | 3.66 |

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# DENELE ANALYTICAL, INC.

865 S. KILROY
TURLOCK, CA 95380
PH# 209-634-9055 FAX# 209-634-9057

## DAIRY COMPLIANCE

Client: CARLOS LOURENCO DAIRY
Mailing address: 6557 Oak Avenue, Merced CA 95341
(not facility address)
Phone#: 209-495-8654

Authorized Copy Released To:
hlcdan@yahoo.com

Sampled By: Daniel Roos
Project ID:

### ANALYSIS TO BE COMPLETED:

**IRRIGATION/GROUND WATER**
H1 EC*, NO3-N, NH4-N*
H6 EC, TN, TDS
H7 Combination of H1 & H6
H14 EC, NO3-N, NH4-N, Ca, Mg, Na, Bicarb Carb, SO4, Cl, TDS

**PROCESS WASTE WATER**
H3 EC*, NH4-N, TKN, TP, K, TDS
H4 EC*, NO3-N, NH4-N, TKN, TP, K, TDS
H5 Ca, Mg, Na, Bicarb, Carb, SO4, Cl
H8 Combination of H3 &H5
H9 Combination of H4 &H5

**TILE DRAIN**
H2 EC*, NO3-N, TP, NH4-N*, TDS

** Required Field Measurement*
___ (Check here if performed)

**DISCHARGE WATER**
E1 EC*, Temp*, pH*, NO3-N, NH4-N, TKN, TP, K, TDS, Dissolved Oxygen*, BOD 5, TSS, Total & Fecal Coliforms
E2 EC,* Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Dissolved Oxygen*, Total & Fecal Coliforms
E3 EC*, Temp*, pH*, NO3-N, NH4-N*, TP, Total & Fecal Coliforms
E4 EC*, Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Total & Fecal Coliform

**MANURE**
M1 % Moisture
M3 TN TP, K, % Moisture
M4 Ca, Mg, Na, Bicarb, Carb, SO4, Cl, Ash
M7 Combination of M3 & M4

**PLANT TISSUE**
F18 % Moisture, Ash, TN, P, K, DM and as received
F18+ % Moisture, Ash, TN, P, K, DM and as received (Package also includes NIR DCAD)
F19+ % Moisture, TN (Package includes complete plant tissue analysis)

**SOIL**
K3 Soluble Phosphorus
K4 1ft NO3, Organic Matter
K5 Pre-Plant (2ft NO3), Fall (1-2ft NO3), Fall (2-3ft NO3)
K6 0-1ft EC, NO3, Soluble P, K, Organic Matter
K7 Combination of K3 & K4
K8 Combination of K3 & K5

| | Date Collected | Time Collected | Sample ID | Description | Temp | Well Purge | H1 | H3 | M3 | F18 | OTHER |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 8-24-23 | 1400 | #2-CS | COLD SILAGE | | | | | | X | |
| 2 | 8-25-23 | 1100 | #3-CS | " | | | | | | X | |
| 3 | 8-25-23 | 1350 | #4-CS | " | | | | | | X | |
| 4 | | | | | | | | | | | |
| 5 | 8-29-23 | 1215 | D-#3 | | 18.0° | RUNNING | X | | | | |
| 6 | 8-29-23 | 1225 | D-1,2 | | 17.0° | " | X | | | | |
| 7 | | | | | | | | | | | |
| 8 | | | | | | | | | | | |
| 9 | | | | | | | | | | | |
| 10 | | | | | | | | | | | |

Analysis

### CHAIN OF CUSTODY RECORDING

| | Signature | Company | Received Date & Time | Relinquished Date & Time | (LABORATORY USE ONLY) Logged In By: | LABORATORY NUMBER |
|---|---|---|---|---|---|---|
| 1st | | HeartLand Consulting | | 8-29-23 1330 | | T3241016 |
| 2nd | | DENELE ANALYTICAL, INC. | 8-29-23 1330 | | | |

Required to be submitted to any agency? (YES / NO)